

Generalized non-*de dicto*: Underspecification and discourse stability*

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Abstract

Embedding material under intensional operators gives rise to a variety of ‘non-*de dicto*’ construals: the proposition seemingly expressed by the embedded clause does not describe the attitudinal state of the subject, yet the sentence can be judged true. This paper makes three contributions: First, we strengthen the claims by Mayr and Schmitt 2024 that non-*de dicto* construals are a uniform phenomenon, which does not involve dedicated syntactic structures or morphology: The distinction between *de dicto* and non-*de dicto* construals does not involve ambiguity but, and this is our second point, arises from properties of meaning *per se*. What can be described as ‘replacement’ – in non-*de dicto* construals the meaning of a constituent of the embedded clause is replaced by another – results from meanings being underspecified: any expression introduces a set of meanings and intensional operators existentially quantify over such sets. Non-*de dicto* phenomena thus form a natural class with other potential instances of underspecification. Finally, as the underspecification we assume is more radical as in underspecification accounts of other phenomena, we propose a new mechanism constraining it, roughly based on pragmatic considerations. Discourse participants aim for ‘discourse stability’: the ‘traditional’ meanings of expressions and their replacements must be indistinguishable for all the questions that might occur in the projected discourse.

1 Introduction

The fundamental property of intensional operators is that they shift the world parameter relative to which some of their arguments are evaluated.¹ Take the attitude predicate *believe*: The embedded clause in (1) describes the world according to Joe, not the ‘actual’ world in which the sentence is uttered, so the semantic interpretation of each expression embedded under *believe* and of the resulting complex expressions is relativized to Joe’s doxastic alternatives (‘belief worlds’). Following Hintikka’s 1969 classical proposal, the sentence is then true just in case it is raining in each such world, i.e., if the proposition **that it is raining** follows from Joe’s belief state.²

(1) *Joe believes that it is raining.*

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¹More realistically, the evaluation parameter is at least a pair of a world and a time. For reasons of space, we focus on the world component, although some of the phenomena we discuss in this paper have also been discussed for times.

²We use *italics* for natural language expressions, and **bold face** for English renderings of their meanings whenever the details of the meanings are not that important. I.e., **Joe** is the meaning assigned to *Joe*.

1.1 A well-known complication

Assume a general setting where Joe went to a party and forms certain beliefs about romantic relationships between the guests. I utter the sentence in (2). My utterance will be true if Joe told me *Eve is involved with a linguist!*. The proposition expressed by the embedded clause, **Eve is involved with a linguist**, adequately describes all of Joe's belief worlds, so (2) is correctly predicted to be true by the classical analysis. We will refer to this expected interpretation as the *de dicto* (DD) construal: only Joe's perspective matters. If we add to our scenario that Eve wasn't at the party, and that Joe mistook a dancer at the party for Eve, (2) will still be judged true.

(2) *Joe believes that Eve is involved with a linguist*

A familiar problem for the classical approach is that it fails to explain why sentences with intensional operators can be judged true when the material embedded under the operator does *not* adequately describe the worlds the operator takes us to. We here subsume all such construals under the term '*non-de dicto* (NDD)'. For example, (2) can also be true if Joe told me: *The person dancing is involved with a linguist*, but, as opposed to us, he isn't aware that the dancer is Eve; he thinks Eve is single and wasn't at the party. This particular case exemplifies the so-called *de re* construal³: While Joe's belief is about a certain individual (the dancer), the identification of this individual with Eve happens only 'outside' of his belief state (via the speaker).

NDD-construals are not limited to *de re*, i.e., 'outside identification' of primitives like individuals, but can also concern descriptive content of embedded material. Under the classical approach, all the expressions in the embedded clause in (2) are relativized to Joe's belief worlds. We thus correctly predict (2) to be true if Joe told me *The three people Eve was talking to are all linguists. I'm certain she's involved with one of them, but I don't know which one*: The proposition expressed by the embedded clause is entailed by Joe's belief state (in each of Joe's belief worlds there is a linguist involved with Eve). The sentence remains true if we add to the scenario that while Joe and I saw the same people talking to Eve, I know they are biologists, not linguists. This is expected, as the embedded clause expresses Joe's description of the world, so embedded descriptive material like *linguist* is construed *opaquely*.

But again, we find cases where the classical view fails to derive acceptability. Assume Joe told me *No idea what three people Eve was talking to. I'm certain Eve is involved with one of them, I just don't know which one*. Then my utterance of (2) can be judged true if we know that the three people are linguists. This is a *transparent* construal of *linguist*: it's not Joe's description, but ours (another instance of 'outside interference'). We are again dealing with an NDD-construal: **Eve is involved with a linguist** is not entailed by Joe's belief state.

In both types of NDD-construals we seem to attribute a proposition to Joe's belief state that is not consistent with it, and in both cases, the problem is tied to a sub-constituent of the embedded clause: In the *de re* case, we use *Eve* although Joe, as opposed to us, does not identify the dancer with **Eve**. Descriptively, if we *replaced Eve* by the concept **the person dancing** in the interpretation of (2), the resulting proposition **the person dancing is involved with a linguist** would follow from Joe's belief state. Likewise, in the transparent case, we use *linguist* although this is our description, not Joe's. Again, if we replaced **linguist** by the property **person Eve talked to** (Joe's description), the resulting proposition **Eve is involved with a person talking to Eve** would be entailed by his belief state. So NDD-construals can be described as in (3):

(3) NDD CONSTRUALS: In a configuration $[\mu [\phi \alpha \beta]]$ a NDD construal of ϕ obtains if the argument supplied to μ is a proposition p that could be gotten by replacing α with γ in ϕ and leaving the rest as in ϕ , and $\llbracket \gamma \rrbracket \neq \llbracket \alpha \rrbracket$.

³Some of the literature uses the term *de re* for all NDD-construals; we do not, for reasons that will become clear.

But why are utterances of sentences under NDD-construals ever judged true? And under which conditions *are* they judged true, which is to say: when can we replace one concept with another?

1.2 Claims of this paper

We will side with the majority of existing works in one respect: Rather than deriving NDD-construals by weakening the semantics of attitudes (see e.g., Blumberg and Lederman (2021)), we assume that we must alter our assumptions about the contribution of the embedded clause.⁴ In most other respects, this paper will depart significantly from the previous literature.

1.2.1 A novel formulation of the familiar problem

Virtually all existing work assumes that NDD-construals are due to dedicated grammatical devices – so NDD-construals are *readings* individuated by the grammar. Most existing work (with the exception of Percus 2021, Tancredi and Sharvit 2022, Benbaji-Elhadad 2023) moreover treats the different types of NDD-construals as independent phenomena that result from different grammatical strategies. While many analyses of *de re* appeal to the notion of replacement (Quine 1956, Lewis 1979, Percus and Sauerland 2003, Ninan 2012, Charlow and Sharvit 2014, Pearson 2015, Deal 2018 a.o.), accounts of transparent construals usually do not, deriving such construals from mechanisms in the structure that let us evaluate the relevant constituent relative to the actual world (Kamp 1971, Percus 2000, Keshet 2011, Schwarz 2012 a.m.o.).

We argue against both assumptions: First, Mayr and Schmitt (2024) claim that empirical parallels between different types of NDD-construals warrant a uniform treatment and that this treatment should involve replacement (see Percus 2021, Benbaji-Elhadad 2023 for related claims). As evidence for the reality of replacement, they cite the fact that NDD-construals are constrained by a condition that requires ‘comparing’ the proposition with replacement and that without relative to the salient question under discussion (QuD). We adopt their view, proposing, however, a different QuD condition that is linked directly to the notion of replacement.

Moreover, we follow and provide more evidence for Mayr and Schmitt’s 2024 claim that NDD-construals are not individuated in the grammar: NDD- and DD-construals do not result from different (morpho-)syntactic structures.⁵ But importantly, we go a step further: Showing that, *contra* claims in the literature, the availability of NDD-construals is not constrained by the syntactic or semantic category of the expression targeted by replacement, we argue that NDD-construals and thus replacement reflect properties of meaning *per se*. So in example (2), the basic interpretation of the expressions in the embedded clause (e.g., *Eve* or *linguist*) will provide the replacements (e.g. **the person dancing** or **person talking to Eve**). Our claim thus is that NDD-construals force us to adopt a novel theory of the meaning of simple and complex expressions.

1.2.2 A novel analysis

We propose that interpretation involves radical underspecification, modeled here via a bidimensional system: The interpretation function assigns each expression a pair of an ordinary meaning and a set of ‘transparency alternatives’ – the latter responsible for underspecification – as sketched in (4) for *linguist*. The two tiers combine independently to form the meanings of complex expressions, resulting in pairs as schematized in (5) for the embedded clause in (2). Parallel composition only stops if the derivation includes an operator sensitive to the transparency value. Intensional

⁴Our claim below about the content of the embedded material will affect the denotation of intensional operators, but not at the level of world quantification.

⁵As we are not dealing with ambiguity, the term ‘construal’ is unwarranted; we still use it for expository purposes.

operators are such expressions, quantifying existentially over the transparency value of their complements.

- (4) $\langle \text{linguist}, \{\text{linguist}, \text{person talking to Eve}, \dots\} \rangle$
- (5) $\langle \text{that Eve is involved with a linguist}, \{\text{that Eve is involved with a linguist}, \text{that Eve is involved with a person talking to Eve}, \dots\} \rangle$

The crucial question then is how underspecification is reined in. Quite obviously, we don't want **believe** in (2) to quantify over arbitrary propositions like **Ada hit a bear**.

We propose that these constraints fall out from more general pragmatic considerations: given the severe underspecification, the speaker must guarantee that they won't lead the hearer 'too far astray'. Borrowing to some extent from accounts of imprecision with other phenomena (in particular Malamud 2012, Križ 2015, Haslinger 2024), which constrain underspecification via indistinguishability relative to an underlying issue, we propose a QuD condition on replacements that encodes the objective that the speaker and the hearer can have a coherent discourse or what we term 'discourse stability'; essentially, the speaker guarantees that whatever the replacement **f** for, say, **Eve** in (2), the 'indistinguishability' of **f** and **Eve** remains constant throughout all issues (QuDs) in the projected discourse. Moreover, as the speaker did not use a salient (syntactic) alternative for *Eve* (say, *Bea*), the hearer can infer that there is no replacement **g** for **Bea** such that the speaker takes the difference between **g** and **Bea** to be immaterial in the same way.

This comprises a major shift in perspective: NDD-phenomena now form a natural class with underspecification phenomena like non-maximality with definite plurals or imprecision in numerals (see e.g., Krifka 1996, Lasersohn 1999, Malamud 2012, Križ 2015, Križ and Spector 2021, Haslinger 2024 for plurals, Krifka 2007, Solt 2014 for numerals), and constraints on NDD-construals are no longer conceptualized via 'proper semantic' relations like co-extensionality or acquaintance (as in virtually all of the existing literature), but as a by-product of more pragmatic constraints on underspecification.

1.3 Structure of the paper

In section 2 we argue that the *de re/de dicto* distinction cannot be due to a grammatical mechanism since *de re* and *de dicto* construals must be available simultaneously. We spell out the first step of our proposal in which NDD-phenomena directly fall out from semantic underspecification. Section 3 provides further support for this idea showing that NDD construals are constrained by semantic or syntactic category. Section 4 then proposed that underspecification constrained by discourse stability. Section 5 concludes the paper.

2 *de re* from underspecification

This section starts by discussing *de re* and *de dicto* construals and by outlining the account of the former in terms of replacement. We then present data showing that replacement in (and thus the semantic derivation of) *de re* cannot be due to a dedicated grammatical structure, as the original concept and the replacement concept must be accessible in parallel, via the same structure ('parallelism'). Hence the *de re/de dicto* distinction is not a case of grammatical ambiguity. From this, we develop an account where the option of replacement is a core feature of meaning assignment *per se*, namely, the result of radical underspecification: expressions do not only introduce an 'ordinary' meaning, but also 'meaning options', and intensional operators quantify existentially over this set of meaning options. This will directly derive the parallelism facts.

2.1 *de re* vs. *de dicto*

The sentence in (6b) is judged true in scenario 1.⁶ On the classical view sketched above, (6b) is true if and only if Bob’s belief state entails the propositional argument of **believe**, so *believe* has a lexical entry like (7), with $\text{DOX}_{x,w}$ the set of x ’s belief worlds in w .

- (6) a. SCENARIO 1: Joe and Bob went to a party. Ann and Eve were among the guests. Bob knows both Ann and Eve well. He didn’t see them together at the party. Still, he is convinced Eve and Ann are a couple. Joe recognizes Eve, but not Ann and does not think Eve knows Ann. He saw them dancing with each other and thinks Eve and the person she was dancing with are dating. There is discussion whether Eve is single.
 b. *Bob thinks that Eve is involved with Ann.* ✓

$$(7) \quad \llbracket \text{believe} \rrbracket^w = \lambda p_{\langle s,t \rangle} . \lambda x_e . \forall w' \in \text{DOX}_{x,w} (p(w'))$$

In (6b), the argument of **believe** can simply be the proposition generally assumed to be the denotation of the embedded clause, (8a), which yields the truth-conditions in (8b), assigning **true** to (6b) in scenario 1. This is the *de dicto* construal of (6b) as (8a) is entailed by Bob’s belief-state.

- (8) a. $[\lambda w'. \text{Eve is involved with Ann in } w']$
 b. $\llbracket (6b) \rrbracket^w = \forall w' \in \text{DOX}_{\text{Bob},w} (\text{Eve is involved with Ann in } w')$

The sentence in (9) is also true in the scenario 1, but the proposition in (8a) is not true in any of Joe’s belief worlds.

- (9) *Joe thinks that Eve is involved with Ann.* ✓ in scenario 1

In contrast, the proposition in (10) is true across Joe’s belief worlds. Thus, by replacing $[\lambda w. \text{Ann}]$ with (10b), (9) becomes true in scenario 1 in virtue of a *de re* construal.

- (10) a. $[\lambda w'. \text{Eve is involved in } w' \text{ with the person dancing with Eve in } w']$
 b. $[\lambda w'. \text{the person dancing with Eve in } w']$

Simplifying somewhat, many accounts of *de re* hold that when (9) is used in scenario 1, Joe has a particular belief about Ann, the so-called *res*. The important point for now is that Joe does so in virtue of a particular individual concept (Quine 1956, Kaplan 1968, Lewis 1979 a.m.o.), (10b). Assume for now that f contextually resolves to this concept: Replacing $[\lambda w. \text{Ann}]$ with f , we obtain the truth-conditions in (11) for (10), and thus truth in scenario 1.

$$(11) \quad \llbracket (9) \rrbracket^w = 1 \text{ iff } \forall w' \in \text{DOX}_{\text{Joe},w} (\text{Eve is involved with } f(w') \text{ in } w')$$

For what follows we can think of the concept **the person dancing with Eve** as being an alternative or replacement concept for the concept **Ann**.⁷

2.2 Parallelism between *de re* and *de dicto*

We now ask how, on a general level, replacements are made available. The point – which follows Mayr and Schmitt 2024 – will be that the *de re/de dicto* distinction is not a case of grammatical

⁶Mayr and Schmitt 2024 argue that the availability of NDD-construals is constrained by factors tied to the QuD. We postpone discussion of this issue to Section 4, but throughout the paper the contexts control for QuD.

⁷We here omit two restrictions usually taken to be operative on replacement (Kaplan 1968, also cf. (Sharvit 1998, Percus and Sauerland 2003) a.o.): co-extensionality (requiring that the value of f in the actual world yields Ann) and acquaintance. We will discuss these constraints in Section 4 and will argue that they are not needed.

ambiguity: We do not have two distinct structures like those schematized in (12)⁸, where one, (12a), only yields the *de dicto* construal, and the other one, (12b), only the *de re* construal. In fact, we will argue that there are no dedicated structures for either construal: If a structure permits a *de re* construal, it also permits a *de dicto* construal, and, crucially, *vice versa*.^{9,10} Thus there is no special *grammatical* device tied to *de re* construals via which replacements are introduced.

- (12) a. [Joe [believes [that Eve is involved with Ann]]] only *de dicto*
 b. [Joe [believes [that Eve is involved with [μ Ann]]]] only *de re*

We start by noting that the sentences in (13b) can be true in scenario 2.

- (13) a. SCENARIO 2: Joe went to a party. Ann and Eve were among the guests. Joe recognizes Eve, but not Ann (and does not think Eve knows Ann). He saw them dancing with each other and thinks Eve and the person he saw her dancing with are lovers. There is discussion whether Eve is single. No other guest has any opinion about it.
 b. (i) *Exactly one guest thinks that Eve is involved with Ann.* ✓
 (ii) *Only one guest thinks that Eve is involved with Ann.* ✓
 (iii) *Only Joe thinks that Eve is involved with Ann.* ✓

The sentences in (13b) are non-monotonic: their meaning has both an upward monotonic and a downward monotonic component, introduced by *exactly* and *only*. The positive components are true in scenario 2 on a *de re* construal, i.e., if Ann functions as the res and [λw . the person dancing with Eve in w] is the alternative concept for [λw .Ann] supplied by replacement. But which property is involved in the negative component? Is it only the *de re* belief? If *de re* and *de dicto* construals were grammatically individuated as in (12), this would be the prediction.

Scenario 2 is not informative on this matter: it is compatible with the negative components of the sentences entailing only that the *de re* construal of the embedded clause is false for any guest other than Joe. But it is also compatible with the negative components entailing that both the *de re* and the *de dicto* construals are false for any other guest – as no guest other than Joe has any belief about Eve’s relationship status. So (13) does not rule out the possibility that the truth-conditions of the positive component amount to the disjunction of the *de re* and *de dicto* construal.

Whether the negative component involves this disjunction can be tested by modifying the scenario so that some guest other than Joe has a *de dicto* belief about Eve’s relationship status in addition to Joe having the *de re* belief. Our scenario 1, repeated in (14a), is a case in point. The sentences from (13b) repeated in (14b) are very odd here, despite the fact that their positive components are met in (14a) for the same reason they were met in (13a). If the negative components only entailed that the *de re* construal of the embedded clause is false for any guest other than Joe, the sentences should be acceptable – contrary to fact.

- (14) a. SCENARIO 1: Joe and Bob went to a party. Ann and Eve were among the guests. Bob knows both Ann and Eve well. He didn’t see them together at the party. Still, he is convinced Eve and Ann are a couple. Joe recognizes Eve, but does not recognize Ann and does not think Eve knows Ann. He saw them dancing with each other and thinks Eve and the person he saw her dancing with are lovers. There is discussion whether Eve is single. No other guest has an opinion on the matter.

⁸ μ just serves to distinguish the structures. We address actual accounts assuming ambiguity below.

⁹See Wang 2025 for different data motivating the same point.

¹⁰As we assume context to have an effect on which replacements can be picked, the generalization should read as: If a structure S permits a *de re* construal in a context c , there is a context c' in which S permits a *de dicto* construal, and if a structure S permits a *de dicto* construal in a context c , there is a context c' in which S permits a *de re* construal. We comment below on a slight asymmetry in availabilities and how it is predicted by the theory to be developed.

- b. (i) *Exactly one guest thinks that Eve is involved with Ann.* ✗
(ii) *Only one guest thinks that Eve is involved with Ann.* ✗
(iii) *Only Joe thinks that Eve is involved with Ann.* ✗

This shows that the negative components of the sentences in (14b) also entail that the *de dicto* construal is false for any guest other than Joe. As this is not the case in scenario 1 (due to Bob's belief), the sentences come out as unacceptable. Accordingly, the positive component of all the sentences in (14b) amounts to the disjunction of the *de re* and the *de dicto* construal.

We also find the reverse situation. The positive component of (15) is true in scenario 1 in virtue of Bob's *de dicto* belief. For (15) to come out as unacceptable in the scenario, its negative component on the *de dicto* construal must also access Joe's *de re* belief. ((15) is slightly less unacceptable in scenario 1 than the sentences in (14b). We address this issue in Section 2.3).

- (15) *Only Bob thinks that Eve is involved with Ann.* ??/✗ in scenario 1

All of this suggests that both the *de re* construal and the *de dicto* construal are available simultaneously, a conclusion supported by the fact that (16) is true in scenario 1. In scenario 1 Joe has a *de re* belief and Bob a *de dicto* one, and no one else has either of the two, so for the sentence to come out true, both the *de re* and *de dicto* construals must be available.

- (16) *Two guests think that Eve is involved with Ann.* ✓ in scenario 1

Following Lewis (1979) and Percus and Sauerland (2003) (a.o.), let us assume that the replacement concept f used in the truth-conditions in (11) above is in fact existentially quantified over¹¹, so that the truth-conditions for (9) – and (6b) – are more accurately represented as in (17).

- (17) $\llbracket (9)/(6b) \rrbracket^w = 1$ iff $\exists f_{\langle s,e \rangle} \forall w' \in \text{DOX}_{\text{Joe/Bob},w} (f(w') \text{ is involved with Eve in } w')$

This suffices to account for the parallelism data in (16). Assume for concreteness that the subject quantifiers in the matrix clause in (14b), (15) and (16) have denotations like in (18).

- (18) a. $\llbracket \text{exactly one guest} \rrbracket$
 $= \lambda f_{et}. \exists X [\text{guest}(X) \wedge |X| = 1 \wedge f(X) = 1] \wedge \neg \exists X [\text{guest}(X) \wedge |X| > 1 \wedge f(X) = 1]$
b. $\llbracket \text{only one guest} \rrbracket$
 $= \lambda f_{et} : \exists X [\text{guest}(X) \wedge |X| = 1 \wedge f(x) = 1] . \neg \exists X [\text{guest}(X) \wedge |X| > 1 \wedge f(X) = 1]$
c. $\llbracket \text{only Joe} \rrbracket = \lambda f_{et} : f(\text{Joe}) = 1] . \forall x [x \neq \text{Joe} \rightarrow f(X) = 0]$
d. $\llbracket \text{only Bob} \rrbracket = \lambda f_{et} : f(\text{Bill}) = 1] . \forall x [x \neq \text{Joe} \rightarrow f(X) = 0]$
e. $\llbracket \text{two guests} \rrbracket = \lambda f_{et}. \exists X [\text{guest}(X) \wedge |X| \geq 2 \wedge f(X) = 1]$

(19a) gives the denotation for the matrix VP for our examples. While it is clear that the existential quantifier over concepts must scope below the subject quantifier, we postpone a compositional derivation of the VP to Section 2.3. The resulting truth-conditions are as in (19b). Leaving distributivity implicit, (19b) is false in scenario 1: there is a guest y other than Joe for whom there is a concept f such that in all of y 's belief worlds w' Eve is involved with $f(w')$ in w' . Namely, for Bob the concept $[\lambda w. \text{Ann}]$ qualifies. The *only*-cases (including (15)) work in parallel and are thus not detailed here. Finally, *two guests* in (16) has the truth-conditions in (19c). These yield **true** in scenario 1 since there are two guests x (Joe and Bob) such that for each of them there is a concept f such that in all of x 's belief worlds w' Eve is involved in w' with $f(w')$.

- (19) a. $\llbracket \text{thought that Eve is involved with Ann} \rrbracket^w$
 $= \lambda x_e. \exists f_{\langle s,e \rangle}. \forall w' \in \text{DOX}_{x,w} (\text{Eve is involved with } f(w'))$

¹¹But cf. Heim 1994, Deal 2018 for treatments without quantification similar to the one above.

- b. $\llbracket (14bi) \rrbracket^w$
 $= 1$ iff $\exists X[\text{guest}(X) \wedge |X| = 1 \wedge \exists f_{\langle s,e \rangle}. \forall w' \in \text{DOX}_{X,w}(\text{Eve is involved with } f(w'))] \wedge$
 $\neg \exists X[\text{guest}(X) \wedge |X| > 1 \wedge \exists f_{\langle s,e \rangle}. \forall w' \in \text{DOX}_{X,w}(\text{Eve is involved with } f(w'))]$
- c. $\llbracket (16) \rrbracket^w$
 $= 1$ iff $\exists X[\text{guest}(X) \wedge |X| \geq 2 \wedge \exists f_{\langle s,e \rangle}. \forall w' \in \text{DOX}_{X,w}(\text{Eve is involved with } f(w'))]$

Summarizing, the parallelism property supports the view that there is existential quantification over alternative concepts used for replacement in *de re* construals, i.e., the value of the concept variable f should not be left up to the context. Moreover, such existential quantification over alternative concepts must always be an option, even when we are dealing with *de dicto* construals.

This strongly suggests that there are no dedicated grammatical structures via which replacements and existential quantification over the latter are introduced.¹² To see the point, let us apply the widesaafvfqaffpread account of *de re* via grammatically represented concept generators (cf. Percus and Sauerland 2003, Anand 2006, Ninan 2012, Charlow and Sharvit 2014, Pearson 2015, Deal 2018 a.o.) to our data. Our truth-conditions for *de re* construals correspond to those derived with concept generators (modulo suitability conditions on concepts). A concept generator maps individuals to individual concepts and thus would thus stand in for the relevant notion of replacement on this approach. Attitude predicates would then existentially quantify over such generators. Accordingly the *de re* LF for the VP *thinks that Eve is involved with Ann* would look like (20a) with G_2 being the concept generator abstracted over. The *de dicto* LF could be based on (20a), but in principle could do without a concept generator, as in (20b).

- (20) a. $[\text{thinks } [\lambda_2 [\text{that Eve is involved with } [G_2 \text{ Ann }]]]]$
b. $[\text{thinks } [\text{that Eve is involved with Ann }]]$

In (14biii) used in scenario 1, the positive component is only true on the *de re* construal, so it would involve (20a). The negative component would therefore also be based on (20a), and both the availability of a *de re* and a *de dicto* construal would be negated, accounting for the degradedness of (14biii). In (15), however, the positive component is true in virtue of a *de dicto* construal. This is compatible with both (20a) and (20b). But if latter were used, the negative component would also be computed based on it, and only the availability of a *de dicto* construal would be negated. Hence (15) should be true in scenario 1, contrary to fact.

The conclusion for the concept generator view – or any account which derives *de re* construals via special grammatical mechanisms (e.g. Abusch 1994, Heim 1994, Sauerland 2014, Cable 2018) – would then have to be that these mechanisms apply obligatorily, regardless of whether a *de re* construal is intended. While descriptively adequate, such a view would seem to be missing a generalization, as parallelism effects generalize to sentences with an arbitrary number of DD and NDD-construed expressions: In scenario 3, Joe has a *de re* belief regarding both Ann and Eve, while Bob has a *de dicto* belief with respect to both of them. (21bi) and (21bii) are unacceptable, while (21c) is fine. This parallels the data discussed above, showing that each *de dicto* construed expression include the *de re* construal and vice versa.

- (21) a. SCENARIO 3: Joe and Bob went to a party. Ann and Eve were among the guests. Bob knows both Ann and Eve well. He didn't see them together at the party. Still, he is convinced Eve and Ann are a couple. Joe did not recognize either (and does not think Eve and Ann know each other). He saw them dancing with each other. So he thinks the two people he saw dancing with each other are lovers. There was discussion about Eve's and Ann's relationship statuses. No other guest has an

¹²On the present view, the *de dicto* construal involves trivial replacement and would thus in a sense qualify as *de re*. We will nevertheless continue to speak of *de dicto* and *de re* construals for descriptive purposes.

- opinion about either.
- | | | |
|----|--|---|
| b. | (i) <i>Only Joe thinks that Eve is involved with Ann.</i> | ✗ |
| | (ii) <i>Only Bob thinks that Eve is involved with Ann.</i> | ✗ |
| c. | <i>Two guests think that Eve is involved with Ann.</i> | ✓ |

We conclude that a different approach to *de re* construals in terms of replacement is needed, which does not rely on individuated LFs for *de re* and *de dicto* construals – rather, both the original concept and the replacement concepts are available on the basis of a single syntactic structure.

2.3 A proposal based on alternatives

We now spell out a proposal that lives up to this challenge, tying the option of replacement directly to how expressions are interpreted: all expressions come with a set of replacements (alternatives). These alternatives are computed in parallel to the ‘ordinary meaning’. We implement this via a two-dimensional system, as e.g. in alternative-based treatments of focus semantics (Rooth 1985, 1992 a.o.). Intensional existentially quantify over these alternatives sets.

This seems to be the most direct approach to the data discussed. Viewing replacements as part of the basic meaning of expressions might seem overly general, permitting replacement (and thus, NDD-construals) for all kinds of expressions but section 3 will show that this prediction is desirable. The system is also very liberal in another respect, imposing hardly any constraints on *what* can act as a replacement. We return to this issue and the required adaptations in section 4.

2.3.1 Alternatives

The first part of the proposal is that expressions α are assigned both an ordinary semantic value, and an alternative (‘transparency’) semantic value.¹³ The interpretation function $\mathcal{F}$ maps expressions to the two values, i.e., meanings are pairs of objects. This gives us (22) for terminals, illustrated in (23) for *Ann*. $\mathcal{F}$ is relativized to an assignment g , a question Q and a context c , but these parameters play no crucial role in this section. We will discuss their impact in section 4.

(22) If α is a terminal node, then $\mathcal{F}^{g,c,Q}(\alpha) = \langle \llbracket \alpha \rrbracket^o, \llbracket \alpha \rrbracket^t \rangle$, where $\llbracket \alpha \rrbracket^t = \{ \beta : \beta \sim_c \llbracket \alpha \rrbracket^o \}$

(23) $\mathcal{F}^{g,c,Q} = \langle \llbracket \text{Ann} \rrbracket^o, \llbracket \text{Ann} \rrbracket^t \rangle$
 $= \langle \lambda w. \text{Ann}, \{ f : f \sim_c \lambda w. \text{Ann} \} \rangle$

The t -value of terminals is determined relative to the o value by a context-sensitive relation $\sim_c$. For any semantic domain D_a , context c , $\sim_{a,c} \in D_a \times D_a$ is a reflexive relation ‘salient-replacement-for’.¹⁴ This notion, as well as the relevant notion of context, will be fleshed out in section 4.

Given a terminal α , $\llbracket \alpha \rrbracket^t$ contains at least $\llbracket \alpha \rrbracket^o$, plus potentially a number of other denotations from the same domain as $\llbracket \alpha \rrbracket^o$. So a lexical expression enters the derivation together with a set of contextually salient alternatives, as illustrated informally in (24) and (25) for a context where at least the concept **the person dancing with Eve** is a salient replacement.

- (24) a. $\llbracket \text{Ann} \rrbracket^o = \lambda w. \text{Ann}$
b. $\llbracket \text{Ann} \rrbracket^t = \{ \lambda w. \text{Ann}, \lambda w. \text{the person dancing with with Eve in } w, \dots \}$

Composition proceeds roughly as familiar from other uses of two-dimensional semantic systems. Our exposition is simplified in two respects: Pairs of semantic values will be taken to have the type of their first coordinate. Moreover, we assume that the language contains two types of expressions:

¹³We assume an intensional system, so both values are intensions. To avoid high types, we simplify, making co-binding of the world variables of the function and the argument in ‘extensional’ constructions part of the FA-rule.

¹⁴We write ‘ $a \sim b$ ’ for $\langle a, b \rangle \in \sim$. We also suppress the type subscript unless this could lead to confusion.

those whose denotations are not sensitive to the t -value of their sister and expressions that *are* sensitive to that value. To avoid technical overload, we will treat the latter syncategorematically.

In the former case, the o - and the t -values of complex nodes are derived in parallel, combining the o - and t -values of their daughters, respectively, (25). Composition of the o -value employs functional application (FA), (26), composition of the t -value a version of pointwise FA. As intensional operators will target the t -values of their sisters, we only introduce ‘extensional’ FA, i.e. the operation that combines denotations that do not manipulate the world parameter.

(25) If α is a node with daughters β, γ , where $\mathcal{F}^{g,c,Q}(\beta)$ is of type $\langle s, \langle a, b \rangle \rangle$ and $\mathcal{F}^{g,c,Q}(\gamma)$ of type $\langle s, a \rangle$, then $\mathcal{F}^{g,c,Q}(\alpha) = \langle FA(\llbracket \beta \rrbracket^o(\llbracket \gamma \rrbracket^o)), \{f : \exists P \in \llbracket \beta \rrbracket^t, X \in \llbracket \gamma \rrbracket^t(f = FA(P(X)))\} \rangle$.

(26) For any $g \in D_{\langle s, \langle a, b \rangle \rangle}$, $h \in D_{\langle s, a \rangle}$, where a is extensional, $FA(g(h)) = \lambda w. g(w)(h(w))$.¹⁵

Assume we are in a context c where non-trivial replacement concepts are salient only for **Ann**. The pair of values output by $\mathcal{F}^{g,c,Q}$ would then be as follows for *that Eve is involved with Ann*:

(27) $\mathcal{F}^{g,c,Q}(\text{that Eve is involved with Ann})$
 $= \langle \lambda w. \text{Eve is involved with Ann in } w,$
 $\lambda w. \text{Eve is involved with Ann in } w, \lambda w. \text{Eve is involved with the person Eve danced with in } w, \dots \rangle$

2.3.2 Intensional operators and truth

Intensional operators are treated as elements targeting the t -value of expressions (in analogy to other applications of alternative semantics, where special elements, e.g., focus sensitive operators, target the alternative value of their complements): under the current proposal, all expressions that license NDD-construals fall into this class.¹⁶

Take *think*: its denotation quantifies existentially over the elements of the t -value of its complement, as in (28). We simplify and assume that intensional operators block alternative projection from their complement, turning the t -value into a singleton, containing only the o -value.¹⁷

(28) $\mathcal{F}^{g,c,Q}(\text{think } S) =$ (to be revised)
 $\langle \lambda w. \lambda x. \exists p \in \llbracket S \rrbracket^t(\forall w' \in \text{DOX}_{x,w}(p(w'))), \{ \lambda w. \lambda x. \exists p \in \llbracket S \rrbracket^t(\forall w' \in \text{DOX}_{x,w}(p(w'))) \} \rangle$

(30) illustrates the effect of (28) when applied to the embedded clause in (29) (i.e., (27)).

(29) *Joe thinks that Eve is involved with Ann.*

(30) $\mathcal{F}^{g,c,Q}(\text{Joe thinks that Eve is involved with Ann})$
 $= \langle \lambda w. \exists p \in \llbracket \text{that Eve is involved with Ann} \rrbracket^t(\forall w' \in \text{DOX}_{\text{Joe},w}(p(w'))),$
 $\{ \lambda w. \exists p \in \llbracket \text{that Eve is involved with Ann} \rrbracket^t(\forall w' \in \text{DOX}_{\text{Joe},w}(p(w'))) \} \rangle$
 $= \langle \lambda w. \exists p \in \{ \lambda w'. \text{Eve is involved with Ann in } w', \lambda w'. \text{Eve is involved with the person}$
 $\text{Eve danced with in } w', \dots \}(\forall w'' \in \text{DOX}_{\text{Joe},w}(p(w''))),$
 $\{ \lambda w. \exists p \in \{ \lambda w'. \text{Eve is involved with Ann in } w', \lambda w'. \text{Eve is involved with the person Eve}$
 $\text{danced with in } w', \dots \}(\forall w'' \in \text{DOX}_{\text{Joe},w}(p(w''))) \} \rangle$

¹⁵Type a being extensional means that the denotation is not a function with domain **W**.

¹⁶If all intensional operators license NDD-construals (an empirical question that is beyond the scope of this paper, targeting the t -value of their complements will be a characteristic feature of this class.

¹⁷This is a simplification as *think* itself could introduce non-trivial alternatives which then combine with the t -value of the complement. A more adequate (if inelegant) rendering would be (i).

(i) $\mathcal{F}^{g,c,Q}(\text{think } S) =$
 $\langle \lambda w. \lambda x. \exists p \in \llbracket S \rrbracket^t(\forall w' \in \text{DOX}_{x,w}(p(w'))), \{ f_{\langle s(et) \rangle} : \exists g(g \in \llbracket \text{think} \rrbracket^t(\exists X(\mathcal{F}^{g,c,Q}(X) = g(f = \llbracket X \rrbracket^o)))) \} \rangle$

As we equates meanings with pairs of semantic values, we need a rule via which to interpret the matrix node in (29): For now, we employ the truth-definition in (31): It simply says that a pair of propositional type is mapped to true in a world w iff its o -value is mapped to 1 in w .

(31) A pair $\langle X, Y \rangle$ of type $\langle s, t \rangle$ is true in a world w iff $X(w) = 1$

For the sentence in (29) this means that we get true in w if the first element of the pair in (30) maps w to 1, i.e., if (32) is 1 in w .

(32) $\llbracket (29) \rrbracket^o = \lambda w. \exists p \in \{ \lambda w'. \text{Eve is involved with Ann in } w', \lambda w'. \text{Eve is involved with the person Eve danced with in } w', \dots \} (\forall w'' \in \text{DOX}_{\text{Joe}, w}(p(w'')))$

The sentence in (29) is now correctly predicted to be true in our scenario 1 from above: The proposition in (32) is true in the scenario in virtue of Joe believing a proposition in the set denoted by the t -value of the embedded clause, i.e., **Eve is involved with the person Eve danced with**.

2.3.3 Deriving parallelism

Let us take a step back and consider the o -value in (33) for the VP of (29):

(33) $\llbracket \text{thinks that Eve is involved with Ann} \rrbracket^o = \lambda w. \lambda x. \exists p \in \{ \lambda w'. \text{Eve is involved with Ann in } w', \lambda w'. \text{Eve is involved with the person Eve danced with in } w', \dots \} (\forall w'' \in \text{DOX}_{x, w}(p(w'')))$

The property in (33) holds of an individual x as long as x believes *at least one* of the elements of the set denoted by the t -value of the respective embedded clause. For (32b) in scenario 1 in (32a), it is the proposition **Eve is involved with the person Eve danced with** in virtue of which the o -value of the whole sentence is true. Analogously, (34) is also correctly predicted to be judged true in scenario 1, and crucially so also on the basis of (33)– but this time in virtue of another proposition in the set, i.e., **Eve is involved with Ann**.

(34) *Bob thinks that Eve is involved with Ann.* ✓ in scenario 1

Given the existential quantification over alternatives to the denotations of the embedded clauses, the o -value of the sentences in (29) and in (34) are so weak that they cover both the relevant DD- and NDD-construals of the embedded clauses. As there are two individuals making (33) true, we correctly predict the judgment for (35) in scenario 1.

(35) *Two guests think that Eve is involved with Ann.* ✓ in scenario 1

The flip-side of this is that (36) is correctly predicted false in scenario 1, and similarly so for our other non-monotonic examples.

(36) *Only Joe thinks that Eve is involved with Ann.* ✗ in scenario 1

Our account thus predicts the parallelism facts, including, as desired, those involving an arbitrary numbers of (N)DD-construed expressions: once an expression contributes non-trivial replacement alternatives these will figure in t -value of the embedded clause accessed by *think*.

Finally, recall that (37) (= (15)) is slightly less bad than (36) in scenario 1. As opposed to (36), (37) only requires ‘trivial’ replacement alternatives (namely, the element corresponding to the o -component) for the requirements of the positive component of the sentence to be met. So if there are contextual constraints on non-trivial replacement alternatives, as we will argue in Section 4, and if speakers accommodate a context where these constraints prevent non-trivial replacements, they could judge (37) true in the scenario. No such option is available for (36).

(37) *Only Bob thinks that Eve is involved with Ann.*

??/✗ in scenario 1

2.4 Interim summary

We assumed that *de re* comes about by a replacement mechanism and argued that this mechanism is not the result of a dedicated grammatical device: considerations of parallelism militate against LF-ambiguity between replacement, i.e., *de re* construals and non-replacement, i.e., *de dicto* construals. Our account takes these points to heart: Replacement is contributed by the interpretation function itself without the help of special grammatical devices. The next section will show, NDD-construals generalize to virtually all kinds of expressions, as expected if the NDD-option is hardwired into the interpretative system. ‘*de re*’ is then simply the descriptive term for an NDD-construal involving an individual (concept) denoting expression. If *de re* were due to a dedicated grammatical device, the generality of the phenomenon would mean that this device would have to obligatorily co-occur with *any* expression embedded under an attitude predicate.

3 Generalized NDD-construals

NDD-construals where constituents other than those denoting individual concepts seem to be affected are kept separate from *de re* in most of the literature.¹⁸ We will now show that these other kinds of NDD-construals exhibit parallelism, just like *de re*. This warrants a uniform treatment: the pattern motivating the availability of replacements as part of meaning *per se* is not affected by the syntactic configuration or the semantic type of the elements in question.

We first address transparent construals of nominal predicates (which we call ‘narrow-scope, transparent’ (NT)). We then turn to transparent construals of verbal predicates, which have been suggested to be non-existent. Finally, we discuss ‘wide-scope, opaque’ (WO) construals where a quantificational determiner is evaluated outside of the attitude but its restrictor within it.

3.1 Narrow scope, transparent construals

The sentence in (38b) is true in scenario 4. This is expected: the embedded clause is standardly taken to denote the proposition $[\lambda w. \text{a linguist in } w \text{ is involved with Eve in } w]$; this proposition in turn is true in all of Bob’s belief worlds. Hence, (38b) is true on the *de dicto* construal of the embedded clause. In order to differentiate this specific *de dicto* construal of (38b) from the NT construal below, we will call it a narrow-scope opaque (NO) construal with regards to *a linguist*: the indefinite takes narrow scope relative to *think*, and *linguist* is evaluated in the belief worlds.

- (38) a. SCENARIO 4: Joe and Bob went to a party. Bob has no idea if there were any linguists at the party, but is convinced that Eve is in a relationship with a linguist. Joe thought all guests were biologists. Except for three linguists called Ann, Bea, and Cate, this was the case. Joe doesn’t know them or their names. He saw each of them dancing with Eve, who he knows. So he thinks Eve and one of them are lovers but is not sure which of them. There is discussion whether Eve is single.
- b. *Bob thinks Eve is involved with a linguist.* ✓

The sentence in (39), however, is also true in scenario 4.

(39) *Joe thinks Eve is involved with a linguist.*

✓ in scenario 4

¹⁸We mentioned the exceptions above: Percus (2021) and Benbaji-Elhadad (2023) use replacements across the board. Tancredi and Sharvit (2022) treat *de re* and transparent construals by what we will call an ‘extensional’ system.

In contrast to the NO-case, this judgement does not follow straightforwardly from standard assumptions: In scenario 4 [$\lambda w.a$ linguist in w is involved with Eve in w] is not true in all of Joe's belief worlds. The description *linguist* is not Joe's, but comes from 'outside'; it holds in the utterance world. This is the NT construal (also known as 'the third reading', Fodor 1970, Bäuerle 1983 a.m.o.). While the predicate *linguist* is not evaluated relative to Joe's belief state, the indefinite must be construed in the scope of the intensional operator: there is no linguist x in the actual world such that Joe thinks x is involved with Eve. As Joe is undecided with regard to which of Ann, Bea and Cate is involved with Eve, the indefinite must vary with the belief worlds.

This last point means that the NT construal is not reducible to a wide-scope transparent (WT) construal, where the entire indefinite is construed outside of the scope of the intensional operator. scenario 5 would verify this WT-construal of (40b), but this construal is strictly stronger than the NT-construal (the sentence would also be true on the NT construal here).¹⁹

- (40) a. SCENARIO 5: Joe went to a party. He thought all guests were biologists. Except for a linguist called Ann, this was the case. Joe doesn't know her name. As he saw Ann dancing with Eve and didn't see Eve dancing with anyone else, he thinks Eve and the person he saw her dancing with are lovers. Eve is single.
- b. *Joe thinks Eve is involved with a linguist.* ✓

3.1.1 NT through replacement

The replacement approach from section 2.3 immediately extends to NT-construals. Instead of replacing an individual concept, replacement now applies to a property. For (39), the transparency value of *linguist* might look as in (41a). Using the compositional procedure and truth-assumption from above, we derive the truth-conditions in (41b), and (41b) is true in scenario 4 by virtue of Joe's belief state entailing **Eve is involved with a person dancing with Eve**.

- (41) a. $\llbracket \text{linguist} \rrbracket^t = \{\lambda w.\lambda x.x \text{ is a linguist in } w, \lambda w.\lambda x.x \text{ is a person dancing with Eve in } w, \dots\}$
- b. $\llbracket (39) \rrbracket^o = \lambda w.\exists p \in \{\lambda w'.\exists x[\text{linguist}(w')(x) \& \text{involved with Eve}(w')(x)], \lambda w'.\exists x[\text{person dancing with Eve}(w')(x) \& \text{involved with Eve}(w')(x)], \dots\}(\forall w'' \in \text{DOX}_{\text{Joe},w}(p(w''))$

3.1.2 Parallelism between NT and NO construals

If NT-construals are derived by replacement, they should exhibit parallelism. This is exactly what we find. Consider first (42). The sentences in (42b) are all true in scenario 6, so the positive components of the non-monotonic examples hold on the NT construal of *a linguist*. If the negative components of the sentences only entail that the NT construal is false for any guest other than Joe, the sentences come out true in the scenario, but of course the same result obtains if they entail that both the NT and NO construals are false.

- (42) a. SCENARIO 6: Joe went to a party. He thought all guests were biologists. Except for three linguists called Ann, Bea, and Cate, this was the case. Joe doesn't know them or their names. He saw each of them dancing with Eve, who he knows. So he thinks Eve and one of them are lovers but is not sure which of them. There is discussion whether Eve is single.
- b. (i) *Exactly one guest thought that Eve is involved with a linguist.* ✓
- (ii) *Only one guest thought that Eve is involved with a linguist.* ✓

¹⁹Several tests involving e.g., other scope-bearing elements in the matrix clause could be employed to check if WT construals exist (see von Fintel and Heim 2011). We will not engage in this discussion.

(iii) *Only Joe thought that Eve is involved with a linguist.* ✓

Crucially, these sentences, repeated in (43b), are odd in scenario 4, repeated in (43a). The positive components are true on the NT construal of the embedded clause, for the same reason as in (42). If the negative components only entailed that the NT construal is false for any guest other than Joe, they should be acceptable in (43a). Since they are unacceptable, we can conclude that the negative component also entails that the NO construal is false for any guest other than Joe. As Bob violates this requirement in (43a), unacceptability follows.

- (43) a. SCENARIO 4: Joe and Bob went to a party. Bob has no idea if there were any linguists at the party, but is convinced that Eve is in a relationship with a linguist. Joe thought all guests were biologists. Except for three linguists called Ann, Bea, and Cate, this was the case. Joe doesn't know them or their names. He saw each of them dancing with Eve, who he knows. So he thinks Eve and one of them are lovers but is not sure which of them. We are discussing if Eve is single; no other guest has an opinion.
- b. (i) *Exactly one guest thought that Eve is involved with a linguist.* ✗
(ii) *Only one guest thought that Eve is involved with a linguist.* ✗
(iii) *Only Joe thought that Eve is involved with a linguist.* ✗

Just like with *de re*, we also find the inverse: The sentence in (44) is not good in scenario 4. The positive component involves an NO belief of Bob's, but the fact that the sentence is not acceptable suggests that Joe's NT belief is also taken into account in the negative component. (43b) and (44) show that there is no difference between the input for NDD and DD construals: the NT construal must also make the NO construal available and, importantly, *vice versa*.

(44) *Only Bob thought that Eve is involved with a linguist.* ??/✗ in scenario 4

Again, paralleling the situation with *de re*, the acceptability of the sentence with the bare numeral in (45) in scenario 4 corroborates the simultaneous accessibility of the NT and the NO-construal: the former must be available when considering Joe, the latter when considering Bob.

(45) *Two guests thought that Eve is involved with a linguist.* ✓ in scenario 4

Finally, the parallelism property generalizes to arbitrary numbers of nominal predicates embedded under *think*: For (46bi) and (46bii) to be true in scenario 7, *psychologist* and *biologist* must both be construed NT relative to Joe's beliefs and both NO relative to Bob's. The sentences exhibit the by now familiar pattern. Each NO construed expression contains each NT construed and the reverse also holds. Again, the acceptability of (46c) reinforces this conclusion.

- (46) a. SCENARIO 7: Joe and Bob went to a party. Joe thought the guests were all linguists. There were six linguists there. The rest were biologists (Ann, Bea, Cate) and psychologists (Daria, Eve, Fran). Joe doesn't know their names. He saw Ann dancing with Daria, Bea with Eve, and Cate with Fran. He comes to think that one these pairs must be a couple but is not sure which of them. Bob wrongly thinks that the six linguists are actually three biologists and three psychologists. For some reason he comes to think that one of the former is involved with one of the latter.
- b. (i) *Only Joe thinks that a psychologist is involved with a biologist.* ✗
(ii) *Only Bob thinks that a psychologist is involved with a biologist.* ✗
- c. *Two guests think that a psychologist is involved with a biologist.* ✓

3.1.3 Consequence for existing accounts

The data just discussed do not only support our take on NDD in terms of a generalized property of the grammar, they also have a direct impact on existing accounts of the NT/NO distinction. To our knowledge, most such accounts assume some form of syntactic or referential ambiguity between NO and NT construals: there are dedicated structures for NT construals to the exclusion of NO construals.²⁰ We will not go into the details and only highlight the kinds of ambiguity assumed. We can divide accounts into intensional ones with evaluation worlds contributed as interpretation parameters and extensional ones with worlds present in the syntactic structure. Among intensional approaches, some based on movement of the transparently interpreted DP with subsequent semantic reconstruction for NT (von Stechow and Heim 2011), others on the optional presence of an *ACTUAL* operator for NT (Kamp 1971, Cresswell 1990), and still others on abstraction over the world parameter by a special operator Λ with movement across that operator for NT (Keshet 2011). Extensional approaches, on the other hand, have evaluation worlds present in the syntax and vary them accordingly for NO and NT construals (Percus 2000, 2020, Schwarz 2012). (47) shows a pair of dedicated structures for NO and NT construals for (39) for each of these theories.

- (47)
- a. **Ambiguity in movement plus reconstruction approaches**
 - (i) [Joe thinks that Eve is involved with [a linguist]] (NO)
 - (ii) [[a linguist] [λ_2 [Joe thinks that Eve is involved with t_2]]] (NT)
 - b. **Ambiguity in *ACTUAL*-operator approaches**
 - (i) [Joe thinks Eve is involved with [a linguist]] (NO)
 - (ii) [Joe thinks Eve is involved with [a [*ACTUAL* linguist]]] (NT)
 - c. **Ambiguity in world-parameter binding approaches**
 - (i) [Joe thinks [Λ Eve is involved with [a linguist]]] (NO)
 - (ii) [Joe thinks [[a linguist] [λ_2 [Λ [Eve is involved with t_2]]]]] (NT)
 - d. **Ambiguity in extensional approaches**
 - (i) [Joe thinks w^* [λ_w Eve is involved w with [a [linguist w]]]] (NO)
 - (ii) [Joe thinks w^* [λ_w Eve is involved w with [a [linguist w^*]]]] (NT)

Our results suggest that none of these approaches is tenable, as they cannot account for the parallelism data: with separate structures for NT and NO construals we do not predict that the negation of a sentence seemingly construed as NT entails the falsity of the NO construal or *vice versa*.

We should also point out that the parallelism data indicate that the interpretative system is intensional, i.e., world variables are not contributed by pronouns as in (47d). If the system were extensional, the parallels between *de re* and NT construals could be implemented in the way outlined above. But an extensional system would overgenerate NT construals, as there would be an additional route to NT, based on the type of LF in (47d). The parallels between *de re* and NT would remain a mystery even with the option of replacement of properties for NT construals.

3.1.4 Non-local replacement

Our account does not only see replacement as a general property of the grammar in the sense that no dedicated devices are needed, but also in the sense that it involves all terminal expressions (and composition rules for non-terminals). It thus contrasts with recent accounts suggesting that the grammar (only) needs to provide replacement at the propositional level (Percus 2021, Benbaji-Elhadad 2023), implementable through a dedicated operator. So, before we return to the broader data pattern, let us make use of data involving NT to motivate our ‘local’ approach.

Proposals in terms of propositional replacements are based on cases like (48). In scenario 8,

²⁰As referential ambiguity contrasts locally bound and locally free variables, it involves syntactic ambiguity.

Joe does not have a *de dicto* belief about Ann. Percus suggests that in order for (48b) to come out true the entire embedded clause must be replaced by a proposition like **AF62 has landed**. On our approach this is not necessary; We could simply replace **Ann** with a concept like **AF62**.

- (48) a. SCENARIO 8: Ann is on flight AF62. Bea and Cate want to know if she arrived. Bea asks Joe, who works at the airport but does not know Ann is traveling, if AF62 landed. He says he thinks it has. Bea to Cate:
 b. *Joe thinks that Ann has arrived.* ✓
 (adapted from Percus 2021)

But can we show that local replacement is not only possible but necessary? Consider (49b): the embedded clause is contradictory, i.e., it denotes the empty set. Yet, the sentence in (49b) does not have trivial truth-conditions – for example, it can be judged true in scenario 9.

- (49) a. SCENARIO 9: The winners of the race are Abe, Bea, and Cate. Joe, who knows their names, but doesn't know that they are high-performing athletes thinks that one of these three came last but is not sure who. Moreover, Dan and Eric came last. Again, Joe who knows their names and mistakenly thinks they are high-performing athletes, thinks that one of them came first. Me:
 b. *Ha! Joe thinks that one of the winners lost.* ✓
 (adapted from Blumberg and Lederman 2021)

Descriptively, we seem to be dealing with an NT-construal, so Joe believes a proposition that involves replacement. Since alternatives are introduced at the level of terminals in our system, we derive a set of non-trivial alternative propositions for the embedded clause in (49b): its *t*-value might be (50), despite the *o*-value of the embedded clause being the empty proposition. The proposition among (50) that Joe might believe could then be **one of Ada, Bea and Carl lost**. This way the acceptability of (49b) is accounted for by the present proposal.²¹

- (50) { λw . one of the winners in *w* lost in *w*, λw . one of Ada, Bea and Carl lost in *w*}

Notice that this allows us to derive a *t*-value for the embedded clause in (51) that differs from the one for (49b). This is desirable, given the non-equivalence of (49b) and (51) in scenario 9. This result obtains despite the fact that their *o*-values are the empty proposition, because, again, replacement happens at the level of the terminals.

- (51) *Joe thinks that one of the losers won.* ✓in scenario 9

If alternatives were introduced at the propositional level, however, we would make an desirable prediction. Assume there was an operator at the CP level in (49b) and (51) that gave us a set of contextually salient alternative propositions for the standard meaning of the embedded clause: We would be looking for alternatives to the empty proposition. While it is conceivable that there are such alternatives (see section 4), they should be the same for (49b) and (51): all contradictory embedded clauses, irrespective of the linguistic expressions within them, should share a *t*-value.

3.2 Transparent construals of verbal predicates

A potential argument *for* an approach to NDD construals in terms of a dedicated grammatical device comes from the literature arguing that transparent construals of properties can only occur in certain syntactic contexts. For example, Percus (2000) argues that while nominal expressions permit transparent construals, verbal expressions do not (see also Schwarz 2012). (52) is a minimal

²¹We will discuss the conditions under which such sentences can be uttered in Section 4.

modification of Percus' example. On the intended reading *brother* in (52b) is construed opaquely. If *Canadian* could be construed transparently, (52b) should be fine in scenario 10, Percus reasons. As it is not, he concludes that the transparent construals is not possible for verbal material.

- (52) a. SCENARIO 10: Ann mistakenly believes that Bert is Joe's brother. She moreover mistakenly believes that Bert is American, while in fact he is Canadian.
 b. *Ann thinks that Joe's brother is Canadian.* X

(52) could be taken as indirect evidence for a grammatical approach – and against our approach – as it could be taken show that for some reason verbal predicates cannot have the relevant device applied to them; e.g., the distribution of concept generators might be restricted.

(52), however, is not all there is to say about NDD construals of verbal predicates (see also Schwager 2011, Sudo 2014). (53b) is perfectly acceptable in scenario 11. Similarly to (52b) *Canadian* would be evaluated transparently, while *Eve's partners* would be evaluated opaquely. Of course, something needs to be said about the contrast between scenarios (52a) and (53a) in their potential to license replacement of **Canadian**. We will do so in Section 4 when discussing the impact of the QuD on the availability of NDD-construals. At any rate, the acceptability of (53b) in scenario (53a) should be sufficient to show that verbal predicates permit replacement.

- (53) a. SCENARIO 11: Joe went to a party. He thought all guests were American. Except for one Canadian called Ann this is true. He saw Eve dancing with exactly one person and thinks Eve and the person she was dancing with are lovers. The person he saw Eve dancing with happens to be Ann. Joe neither knows Ann nor her name. He also was not able to see Ann's face. We know that Eve changes partners constantly and has multiple ones at any point. Unsure who her current partners are, we are wondering where they might come from. Me:
 b. *Joe thinks that one of Eve's partners is Canadian.* ✓

If verbal predicates allow for transparent construals, they should exhibit the parallelism property as well. Consider the extension of scenario 11 in (54a). (54b) is acceptable here in virtue of an opaque construal of *Canadian*. Crucially, all of (54c) are unacceptable in (54a), while (54d) is acceptable. This is as expected given the parallelism constraint on NDD-construals: the transparent construal of *Canadian* includes the opaque one and vice versa.

- (54) a. SCENARIO 11A: [...] Bob was also at the party. He has been convinced for a long time that one of Eve's many partners is Canadian. He suspects that person of claiming to be American because she has ambitions to run for president.
 b. *Bob thinks that one of Eve's partners is Canadian.* ✓
 c. (i) *Only Joe thinks that one of Eve's partners is Canadian.* X
 (ii) *Only Bob thinks that one of Eve's partners is Canadian.* X
 (iii) *Exactly one guest thinks that one of Eve's partners is Canadian.* X
 d. *Two guests think that one of Eve's partners is Canadian.* ✓

All of this suggests that verbal predicates can receive transparent construals just like nominal ones and, crucially, in the same way. This is as expected on the present view.

3.3 Transparent construals of quantificational determiners

Quantificational determiners can also undergo replacement. We refer to the phenomenon lending support to this conclusion as 'wide scope, opaque' (WO) construals (also termed the 'fourth

reading’, Fodor 1970).²² Descriptively, WO construals are the inverse case of NT construals: the functional material (numerals or determiners), seems to not be evaluated relative to the intensional operator (‘wide scope’), while the descriptive material (the nominal predicate) is construed ‘opaquely’ relative to this operator.²³

Consider (55), adapted from Szabó 2010: The sentence in (55b) is true in scenario 12, so *terrorist* must be evaluated in Abe’s belief worlds but *most* must not: The sentence reports the proportion of people identified as terrorists living on Abe’s street relative to individuals identified as terrorists not living on this street from Bea’s perspective. Abe has no belief regarding such a proportion: The proposition **most terrorists live on Abe’s street**, corresponding to the *de dicto* construal of the embedded clause, is not entailed by Abe’s belief state.

- (55) a. SCENARIO 12: Abe falsely thinks his neighborhood is full of terrorists. He goes to the police. Officer Bea shows him photos of people. For each photo, she asks if the person on it is a terrorist and if they live on Abe’s street. When done, Abe is asked if there are terrorists in the neighborhood who are not on any of the photos, if he knows how many people he identified as terrorists and how many of those as living on his street. Abe says no and claims he only knows he identified several as terrorists living on his street. Bea knows that Abe has identified 11 as terrorists in total and 7 of these as living on his street. Bea tells her colleague, who, like Bea, is worried about Abe going crazy. Bea:
- b. *Abe believes that most terrorists live on his street.* ✓

The replacement theory of NDD is applicable here as well. For instance with the *t*-value for *most* in (56a), i.e., including the alternatives **most** and **several**, and no other expressions contributing alternatives beyond their *o*-value, the truth-conditions of (55b) would amount to (56b). (56b) is true in scenario 12 in virtue of **several terrorists live on Abe’s street** being entailed by Abe’s belief state. We thus derive the truth of (55b) without unorthodox scoping operations (see Szabó (2010) for a proposal along such lines and Benbaji-Elhadad (2023) for criticism).²⁴

- (56) a. $\llbracket \text{most} \rrbracket^t = \{ \lambda w. \lambda f. \lambda g. |f \cap g| > \frac{1}{2}|f|, \lambda w. \lambda f. \lambda g. |f \cap g| > 2, \dots \}$
- b. $\llbracket (55b) \rrbracket^o = \lambda w. \exists p \in \{ \lambda w'. | \lambda x. \text{terrorist}(w')(x) \cap \lambda x. \text{live on A's street}(w')(x) | > \frac{1}{2} | \lambda x. \text{terrorist}(w')(x) |, \lambda w'. | \lambda x. \text{terrorist}(w')(x) \cap \lambda x. \text{live on A's street}(w')(x) | > 2, \dots \} (\forall w'' \in \text{DOX}_{\text{Abe}, w}(p(w''))$

If this is on the right track, WO construals should exhibit parallelism. scenario 13 is parallel to scenario 12 with regard to Abe: (57bi) is true in virtue of a WO construal of **most terrorists**. For the acceptable (57bii) to be true in the scenario *most terrorists* must receive a NO construal, because in scenario 13 Bob has a belief about the proportion between people identified as terrorists living on his street relative to those identified as terrorists not living on his street: he believes that the former group constitutes the majority. Since this is false, **most** must be evaluated relative to Bob’s belief worlds and not Bea’s. Our generalized NDD approach predicts that a WO construal of the embedded clause in (57b) includes the NO one (57a) and vice versa. This is confirmed by the unacceptability of the examples in (57c) and the acceptability of (57d).²⁵

²²The status of these construals as separate readings has been debated in the literature (see e.g., Fodor 1970, Keshet and Schwarz 2019, Benbaji 2021). This issue will be addressed in Section 5; it is immaterial for our present discussion.

²³Our use of ‘scope’ should not be taken to reflect a structural property. On the present account the numerals and determiners in WO-construals, just like the nominal predicates in NT-construals, do not take wide scope syntactically. All we mean by ‘wide scope’ is that the functional material is not evaluated in the worlds quantified over by the attitude.

²⁴WO construals are also possible with numerals, *many*, *few*, etc. Kusliy and Vostrikova (2018) suggest a *de re* analysis for related cases with numerals. On their view this requires numerals to denote entities or degrees. Since our analysis covers all determiners we do not have to make particular assumptions about the denotations of numerals.

²⁵Descriptively, the WO-NO-distinction is one in terms of scope. We argued that the structure supporting WO

- (57) a. SCENARIO 13: Abe and Bob live in the same neighborhood. They both falsely think it is full of terrorists. Both go to the police. Officer Bea shows them photos of people. For each photo, she asks them if the person on it is a terrorist and if they live on their respective street. When they are done, they are asked if there are terrorists in the neighborhood who are not on any of the photos, if they know how many people they identified as terrorists and how many of those as living on their streets. Bob says no and claims he identified 7 terrorists and of these 5 as living on his street. Bea knows that Bob identified 7 photos as showing terrorists and 3 of them as living on his street. Abe also says no, and says he only knows he identified several as terrorists living on his street. Bea knows that Abe has identified 11 as terrorists in total and 7 of these as living on his street. Bea tells her colleague, who, like Bea, is worried about Abe and Bob going crazy:
- b. (i) *Abe believes that most terrorists live on his street.* ✓
(ii) *Bob believes that most terrorists live on his street.* ✓
c. (i) *Only Abe believes that most terrorists live on his street.* ✗
(ii) *Only Bob believes that most terrorists live on his street.* ??/✗
d. *Abe and Bert believe that most terrorists live on their streets.* ✓

3.4 Consequence: No DD/NDD ambiguity

Summarizing, we have evidence for replacement of DPs, NPs, verbal predicates, and quantificational determiners. They all exhibit parallelism and parallelism extends to an arbitrary number of expressions construed NDD/DD. On the concept generator view – our example for a grammatical view of replacement – this would mean that all of the expressions mentioned obligatorily have a generator next to them. That is, the sentence in (58) would need to have an LF like (59).

(58) *Abe thinks that most terrorists live on Bea's street.*

(59) [Abe thinks [[G₁ most] [G₂ terrorists]] [G₃ live] [on [[G₄ Bea's] [G₅ street]]]]

While this would be formally sound, we take it that an approach like the one proposed here where generalized NDD construals fall out as a byproduct of semantic interpretation is to be preferred.²⁶

4 Restrictions on replacement: discourse stability

We argued that permitting replacement of any syntactic and semantic category is a positive feature of our account. But the proposal is also very liberal in terms of which replacements are

must also support NO, which can be captured by assigning a liberal meaning to the former. This raises the question whether any of the classical scope ambiguities should be treated on a par. This option would only be motivated if such configurations also exhibited parallelism. We leave the investigation of this matter to future research.

²⁶Romoli and Sudo (2009) suggest that a DP embedded in another DP cannot be construed opaquely if the latter is construed transparently. We contest this conclusion on the basis of (ib) and (ic) in scenario (ia): *sister* is construed transparently relative to Ann's beliefs, while *guest of honor* is construed opaquely. The sentences are acceptable, suggesting that Romoli and Sudo's generalization does not hold. This is important for us as it rules out another candidate for a syntactic constraint on NDD-construals that could be taken to support a grammatical approach.

- (i) a. SCENARIO: Ann is confused. I have repeatedly told her that Bob is the guest of honor at the party, but she still thinks the party is in Joe's honor. At the party she meets Joe who is there with his sister. Ann tells me that this woman she just saw with Joe must be his mistress and that the look in that woman's eyes shows she wants to kill Joe for his money. I tell the others: *Can you believe how crazy Ann is?* ...
b. *Ann thinks that the sister of the guest of honor is out to kill him.* ✓
c. *Ann thinks that the guest of honor's sister is out to kill him.* ✓

allowed. We will now show that this latter feature makes unwarranted predictions and will thus refine the proposal by constraining replacement. However, we will refrain from reining it in via ‘core semantic’ properties like (co-)extension in the actual world, which, despite the different perspectives, feature both in traditional analyses of *de re* and of NT-construals. Such a view will not only be shown to be incompatible with the generality of the phenomenon, but would also sit awkwardly with the conceptual direction of our account, which appeals to something akin to underspecification at the level of basic meaning assignment. Arguably, in analogy to observations that have been made concerning other cases of underspecification, restrictions should appeal to more ‘pragmatic’ properties, including what is a stake in the conversation.

4.1 Why the current account is too liberal

So far our system allows an expression to contribute any replacement alternative as long as it is contextually salient and of the same semantic type. In this light, consider (60). scenario 14 differs from scenario 1 in that another individual – Bea, who is tangential to Joe’s beliefs – is introduced and thus salient. (60b) is true (as expected). Crucially, and unexpectedly for us, (60c) is not.²⁷

- (60) a. SCENARIO 14: Joe went to a party. Ann and Eve were among the guests. Joe recognizes Eve, but not Ann (...) He saw them dancing with each other and thinks Eve and the person she was dancing with are dating. Bea was also there. Joe does not know her. He saw her talking to the host. There is discussion whether Eve is single.
 b. *Joe thinks Eve is involved with Ann.* ✓
 c. *Joe thinks Eve is involved with Bea.* ✗

On our view, the question is: What allows **the person dancing with Eve** to serve as a replacement concept for **Ann** in (60b) but blocks this option for **Bea** in (60c)?

Could the source of this problem be that our proposal, as opposed to many previous ones, does not require the res and its replacement to be co-extensional in the actual world (Quine 1956, Kaplan 1968, Lewis 1979 a.m.o.)? If it did, the contrast would follow, as **Ann** is co-extensional with **the person dancing with Eve** in the actual world, while **Bea** is not.

The issue also occurs with expressions of other semantic types. scenario 15 differs from scenario 3 in that Joe thinks the guests are all biologists or physicists instead of just biologists. (61b) is true in scenario 15 on the NT construal, but (61c) is not.²⁸

- (61) a. SCENARIO 15: Joe went to a party. He thought the guests were made up of biologists and physicists but could not tell who were the former and who the latter. Except for three linguists called Ann, Bea, and Cate, everyone was indeed either a biologist or a physicist. Joe doesn’t know the three linguists or their names. He saw each of them dancing with Eve, who he knows. He thinks Eve and one of them are dating but is not sure which of them. There is discussion whether Eve is single.
 b. *Joe thinks Eve is involved with a linguist.* ✓
 c. *Joe thinks Eve is involved with a physicist.* ✗

The contrast in (61) parallels the one in (60) for *de re*. But in this case we cannot require the property denoted by the expression to be replaced and its replacement alternative to be co-extensional in the actual world: the set of linguists is not congruent with the set of people dancing with Eve.²⁹

²⁷We thank Nina Haslinger and Mitya Privoznov (pc) for discussion on this issue.

²⁸(61c) is not true on an NO construal either because Joe does not believe that all guests are physicists.

²⁹Implicitly restricting the extension of **linguist** in (61b) to those linguists present at the party would not resolve the issue discussed immediately below in the text.

As the set of linguists overlaps, with that of people dancing with Eve, while the set of physicists does not, we could instead impose a condition in terms of overlap. (For individual concepts, overlap could reduce to co-extensionality.) This option also faces problems: scenario 16 differs from scenario 15 only in that Cate is a physicist, not a linguist, so the set of people dancing with Eve overlaps with both the set of linguists and that of physicists. A salient-replacement-for constraint based on overlap would thus yield **person dancing with Eve** as a possible replacement for both **linguist** in (62a) and **physicist** in (62b). But neither (62b) nor (62c) is acceptable.³⁰

- (62) a. SCENARIO 16: ... Except for two linguists called Ann and Bea, everyone was indeed either a biologist or a physicist. Joe saw each of Ann and Bea and moreover Cate – a physicist – dancing with Eve ...
 b. *Joe thinks Eve is involved with a linguist.* X
 c. *Joe thinks Eve is involved with a physicist.* X

So we might entertain yet another modification. Maybe it is not overlap that matters for replacement but rather the subset/part-of relation. After all, the people dancing with Eve constitute a subset of the linguists but not of the physicists in (61a) and of neither in (62a). This modification would, however, run into problems with our WO examples among other things: in the crucial example in (55) **most** had to be replaced by **several** or a similar quantificational meaning. The extension of the latter is not generally a subset of the extension of the former, and certainly not in the case of (55). Here overlap is the most we can require. This conclusion extends to other cases of WO construals in the literature, for example those with numeral quantifiers.

In summary, our proposal faces an overgeneration problem and the seemingly obvious remedy – imposing constraints on the relation between the extension of the expression used and the value of the replacement concept in the actual world – is not straightforward once we accept that replacement underlies all instances of NDD and also targets expressions with set-type denotations.

4.2 Refining the proposal: discourse stability

We hypothesize that the data just discussed reveal what should be conceived of as a more general constraint on underspecification grounded in the cooperative use of language, i.e., pragmatics. Our underlying intuition is that speakers navigate a system with underspecification by trying to not lead hearers too far astray when making an utterance – and that hearers count on speakers acting in this way. For illustration of this general idea, consider constraints on the imprecise use of bare numerals. While the bare numeral in the sentence in (63) easily lends itself to an imprecise interpretation, a cooperative speaker will not use (63) when they know that Ann owns 420 books.

- (63) *Ann owns 500 valuable books.*

For our immediate concerns this means that speakers will count on hearers ‘knowing’ to only pick those replacements for a sentence like (64) that will avoid certain types of false conclusions.

- (64) *Joe thinks Eve is involved with Ann.*

Such a view would provide a remedy for some strange consequences of our account: Given the option of replacement, by uttering something like (64) we are making a very weak statement. So how can we be informative, i.e., adhere to the maxim of quantity? Relatedly, the notion of quality will be almost meaningless if all speaker are committing to is a weak existential statement.

³⁰Standard approaches to NT construals, which usually do not extend to *de re* construals, derive the contrast between (61) and (62): *Joe thinks that Eve is involved with a linguist* on the NT construal would say that Joe thinks Eve is involved with someone who is a linguist in the actual world.

In our system (and potentially any system employing underspecification), these notions would have to be part of a general principle we call ‘discourse stability’, reflecting the speaker’s goal (or a goal hearers take them to have) that the speaker and the hearer can have a coherent discourse. The first ingredient, based loosely on some conception of quality, would be that the speaker guarantees that whatever replacement **f** the speaker chooses for, say, **Ann** in (64), the ‘indistinguishability’ of **f** and **Ann** will remain constant throughout the projected discourse (or all the issues that might come up). The co-extensionality requirement standardly assumed to hold for *de re* could be seen as a particular instantiation of this more general principle: the latter will automatically be fulfilled if **f** and **Ann** are co-extensional in the evaluation world of evaluation. Discourse stability, however, would allow for more leeway than a simple co-extensionality requirement.

This is where the second ingredient, loosely tied to a conception of quantity, becomes important: by choosing *Ann* over some other expression, say, *Bea*, the speaker indicates that we find no similar innocuous replacement **g** for **Bea**: When the speaker uses *Ann* in (64), they can have **the person dancing with Eve** in mind as a replacement for **Ann** if they think that for all purposes of the projected discourse – what the speaker believes the discourse will be like – **Ann** and **the person dancing with Eve** will be indistinguishable, meaning that the differences between them will be immaterial. And since the speaker did not use a salient (syntactic) alternative for *Ann* (namely, *Bea*), the hearer will infer that there is no replacement **g** for **Bea** such that the speaker takes the difference between **g** and **Bea** to be immaterial throughout the projected discourse.

We here provide a more local implementation of this general intuition,³¹ based on the version of discourse stability in (65): By asserting a sentence, taken to address a QuD, a speaker guarantees that a particular replacement for an expression in it does not resolve this QuD or a projected one differently than the sentence without replacement. The replacement is guaranteed to be innocuous in this sense.

- (65) **Discourse stability (informal)**: Use of a replacement **f** for an expression α is licensed if there is no projected question Q in c s.th. a proposition based on **f** resolves Q differently in c than the same proposition based on $\llbracket \alpha \rrbracket^c$.

We will implement (65) via a definedness condition (‘QuD-constraint’) contributed by attitude predicates and other modal expressions. This narrow implementation will let us deal with embedded occurrences of underspecification and keep the rest of our approach as is.³²

4.2.1 Ways of addressing and resolving a question

Let us clarify the underlying concepts for the discussion of (65)– and the contrast between resolving a question differently and not differently. First, the notion of question relevant for QuDs is in terms of a set of equivalence classes (= cells) of possible worlds, so a question is a partition of logical space **W**, (Groenendijk and Stokhof 1984).

Broadly following Groenendijk and Stokhof (1984), for a question to be resolved by a proposition in a context, the contextual information together with the proposition must entail a cell in the question, as defined in (66a) (cf. Lewis 1988 for the related but different notion of relevance). It must, however, also ‘contribute information’, meaning the context cannot already entail an an-

³¹Turning this rough idea into a full pragmatics would need to adapt the maxims of quantity and quality so as to be applicable in a system involving underspecification. It could be tied to recent work on underspecification with definite plurals or round numerals, where relativized notions of truth (Križ 2015 a.o.) and notions like inference preservation (Haslinger 2024) have been entertained, but our data differ from other cases of imprecision as they not provide a readymade measure for when an imprecise use counts as too distant.

³²A more direct appeal to quantity and quality might let us separate the definedness condition from the contribution of the attitude predicate and the former via an extra operator. One could for instance probe the relation to the presuppositional version of the exhaustification operator, PEX (Bassi et al. 2021).

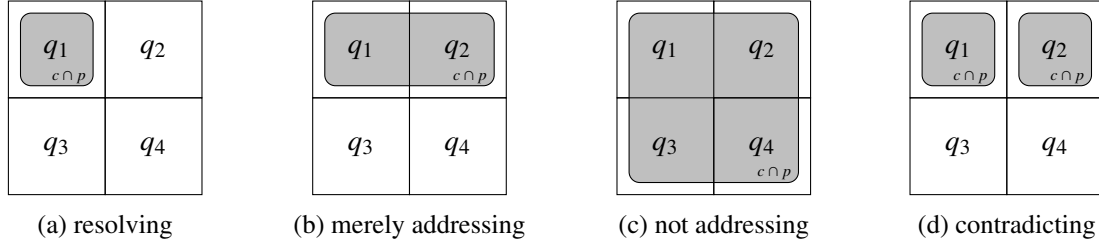


Figure 1: Ways a proposition p can relate to a question Q in context c .

swer (cf. Roberts 2012). Not resolving the question amounts to the complement of this situation. However, we can distinguish different ways of not resolving a question. First, the proposition together with contextual information can address the question without resolving it. This situation obtains when a union of two or more cells is entailed, but no single cell is, (66bi). Conversely, a question remains unaddressed if no union of cells is entailed, (66bii). Finally, a proposition together with contextual information will contradict a question if more than one cell is entailed, (66biii). While formally the question could be said to be addressed in this situation, the outcome would amount to an impossible state since partition cells are by definition incompatible with each other. This situation only obtains if the proposition together with the context is contradictory, so that normally all cells will be entailed. Figure 1 illustrates these four options.

- (66) **Relating to a question in a context:** Given context c , question Q and proposition p ,
- a. p resolves Q in c iff for some $q \in Q$, $c \cap p \subseteq q$ and $c \not\subseteq q$, otherwise
 - b. p does not resolve Q in c , where
 - (i) p merely addresses Q in c , iff for some X such that $X \subset Q$, $c \cap p \subseteq \bigcup X$, and there is no $q \in Q$, $c \cap p \subseteq q$,
 - (ii) p does not address Q in c , iff for no X such that $X \subset Q$, $c \cap p \subseteq \bigcup X$, or
 - (iii) p contradicts Q in c , iff there are $q, q' \in Q$ such that $q \neq q'$ and $c \cap p \subseteq q$ and $c \cap p \subseteq q'$.

Based on the definition in (66), the notions of addressing a question in the same way and of resolving it differently or not differently can be made more precise. We say that two propositions together with contextual information address a question differently if they entail non-overlapping subsets of the question, (67a). Every other conceivable option is a case of not addressing the question differently. This includes situations in which the question is resolved in the same way since the same cell is entailed by the propositions together with the context, (67bi), situations in which the question is addressed in the same way because the same union of cells is entailed without there being agreement on a particular cell, (67bii), and situations in which the question is unaddressed by both propositions. Examples for these different possibilities are given in figure 2.

- (67) **Addressing questions (not) differently:** Given context c , question Q and propositions p and p' ,
- a. p and p' address Q differently in c , iff for some X and X' such that $X, X' \subset Q$, $X \neq X' \neq \emptyset$, $X \cap X' = \emptyset$, $c \cap p \subseteq \bigcup X$ and $c \cap p' \subseteq \bigcup X'$.
 - b. p and p' do not address Q differently in c , $p \sim_c^Q p'$, iff
 - (i) both p and p' resolve Q in c and for some $q \in Q$, $c \cap p \subseteq q$ and $c \cap p' \subseteq q$ (in that case p and p' resolve Q in the same way in c , $p =_c^Q p'$), or
 - (ii) both p and p' address Q in c and for some $X \subset Q$, $c \cap p \subseteq \bigcup X$ and $c \cap p' \subseteq \bigcup X$, (in that case p and p' address Q in the same way in c), or
 - (iii) p and p' do not address Q in c .

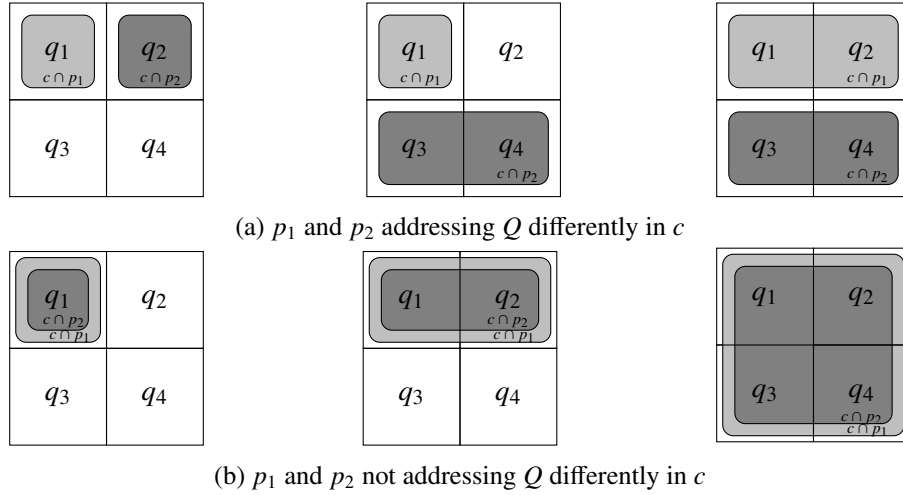


Figure 2: Ways for propositions p_1 and p_2 to jointly relate to question Q .

4.2.2 An asymmetry between utterances relative to replacements

With this background, we return to (60), i.e., the contrast between (68a) and (68b) in scenario 14.

- (68) a. *Joe thinks Eve is involved with Ann.* ✓in scenario 14
 b. *Joe thinks Eve is involved with Bea.* ✗in scenario 14

Notice first that the ordinary values of the embedded clauses in (68a) and (68b) and their alternatives where **Ann** and **Bea** are replaced with **the person Eve danced with** all negatively resolve the QUD ‘Is Eve single?’, which is salient in scenario 14, as shown in (69). Thus they all resolve the QUD in the same way as defined in (67bi).

- (69) a. Salient QUD: *Is Eve single?*
 b. (i) $c \cap \lambda w.$ Eve is involved with Ann in w no
 (ii) $c \cap \lambda w.$ Eve is involved in w with the person dancing with Eve in w no
 $\rightsquigarrow$ resolved in the same way (67bi)
 c. (i) $c \cap \lambda w.$ Eve is involved with Bea in w no
 (ii) $c \cap \lambda w.$ Eve is involved in w with the person dancing with Eve in w no
 $\rightsquigarrow$ resolved in the same way (67bi)

Now take a QUD like (70a), arguably a natural follow-up in scenario 14 when the question whether Eve is single has been resolved negatively. The context contains the information that Ann is the person Eve danced with. If this information is used to evaluate the impact of the embedded clause ‘directly’ relative to (70a), the ordinary value of the embedded clause in (68b) and the alternative based on the replacement **the person Eve danced with** are contextually equivalent. So both resolve the QuD in (70a) positively (and thus in the same way) in the context. Again assuming that contextual information is available when considering the relation of the embedded clause to the QuD, we obtain a different result when comparing the ordinary value of the embedded clause in (68c) and that same alternative, since Bea is not the person Eve danced with: they address the QUD in (70a) differently according to (67a). Thus there is an asymmetry between (68b) and (68c) relative to the QUD ‘Is Eve in a relation with Ann?’, as summarized in (70).

- (70) a. Additional salient QUD: *Is Eve in a relation with Ann?*
 b. (i) $c \cap \lambda w.$ Eve is involved with Ann in w yes

- (ii) $c \cap \lambda w.$ Eve is involved in w with the person dancing with Eve in w yes
 $\rightsquigarrow$ resolved in the same way (67bi)
- c. (i) $c \cap \lambda w.$ Eve is involved with Bea in w not addressed
- (ii) $c \cap \lambda w.$ Eve is involved in w with the person dancing with Eve in w yes
 $\rightsquigarrow$ addressed differently (67a)

This asymmetry would persist if the follow-up QuD were the one in (71b). The ordinary value of the embedded clause in (68a) and the relevant replacement alternative leave it unaddressed in the context and thus do not resolve it differently according to (67b). The ordinary value of the embedded clause in (68b), however, resolves it positively. It and its alternative therefore resolve the question differently following (67a).

- (71) a. Additional salient QUD: *Is Eve in a relation with Bea?*
- b. (i) $c \cap \lambda w.$ Eve is involved with Ann in w not addressed
- (ii) $c \cap \lambda w.$ Eve is involved in w with the person dancing with Eve in w not addressed
 $\rightsquigarrow$ not addressed differently (67biii)
- c. (i) $c \cap \lambda w.$ Eve is involved with Bea in w yes
- (ii) $c \cap \lambda w.$ Eve is involved in w with the person dancing with Eve in w not addressed
 $\rightsquigarrow$ addressed differently (67a)

In light of the notion of discourse stability in (65), we can understand this asymmetry as follows: a speaker who utters (68a) when the QuD is ‘Is Eve single?’ can guarantee that the ordinary value of the embedded clause and the alternative based on the replacement **the person Eve danced with** will not make a difference regarding this question but also regarding follow-up (or projected) QuDs like ‘Is Eve in a relation with Ann?’ and ‘Is Eve in a relation with Bea?’ – as the ordinary value and the alternative will at the least not resolve these projected questions differently. But the speaker could not guarantee this if they were to utter (68b) relative to these same questions.

Note that we assume that utterances of sentences with intensional operators like *think* can relate to questions that do not directly involve the modal operator: For a sentence like (68a) we can check how the content of the embedded clause (instead of that of the entire sentence) relates to the QuD. Such a sentence can thus relate to a question that is not about the subject’s belief state (like ‘What’s Joe’s state of mind’), but can be used in contexts where the issue is connected to the embedded proposition – which also means that the embedded proposition cannot be entailed by the context.³³ It seems to us that this might even be the more typical use. Of course we also find uses (and comment on them below) where the embedded content is entailed by the context (e.g., if we all know that Eve is single in a scenario like 15) and the only plausible QuD that can be addressed would be one targeting the subject’s mental state.

4.2.3 The QuD-constraint

We implement this intuition via the QuD-constraint, a definedness condition imposed by attitude predicates and modals. We proceed in two steps, reflecting the two types of questions that sentences with *think* or intensional operators more generally can relate to: those appealing to the content of the embedded clause, and those appealing to the entire modal statement.

We posit Q_c^S as an evaluation parameter, crucial for the interpretation of *think* and its complement S . Q_c^S is the set of projected questions based on a sentence S and a context c : a set of QuDs

³³We deviate from implicit assumptions in the literature by keeping the notion of addressing a QuD distinct from that of context change: Assessing the embedded content relative to the QuD does not mean adding it to the context.

that is restricted by the structure and content of S and moreover only contains questions that are contextually salient in c . The definedness condition requires that all replacement alternatives of S resolve some QuD Q in Q_c^S in the same way as S in c , following (67bi), and all other QuDs in Q_c^S are *not addressed differently* in c by them, following (67b). The latter means that the questions are either resolved in the same way, or merely addressed in the same way, or not addressed by either of them. The new entry for *think* looks as follows:

$$(72) \quad \mathcal{F}^{g,c,Q_c^S}(\text{think } S)(\langle a, \{a\} \rangle) \\ \text{defined if } \forall p \in \llbracket S \rrbracket^t \rightarrow \exists Q \in Q_c^S (p =_c^Q \llbracket S \rrbracket^o) \wedge \forall Q' \in Q_c^S (p \sim_c^{Q'} \llbracket S \rrbracket^o) \quad (\text{preliminary}) \\ \text{if defined, } \mathcal{F}^{g,c,Q}(\text{think } S)(a) = \langle \lambda w. \exists p \in \llbracket S \rrbracket^t (\forall w' \in \text{DOX}_{a,w}(p(w')), \{ \lambda w. \exists p \in \llbracket S \rrbracket^t (\forall w' \in \text{DOX}_{a,w}(p(w')) \} \rangle$$

The first conjunct of the definedness condition could be seen to be motivated by considerations of quality. It requires that the ordinary value and its alternative make the same contribution regarding some salient QuD. The second conjunct could be tied to considerations of quantity: for all other salient QuDs the ordinary value and its alternative do not make a difference given the context.

Importantly, while the definedness condition relates to the ‘outer’ context – we, the interlocutors, check the status of the embedded content relative to what we are interested in – the assertive component does not. So (72) says that we ascribe a belief to the attitude subject from a set of propositions we consider to have the same status relative to what is at issue, without making the outer context or the indistinguishability requirement part of the subject’s belief.

Assume that the QuDs in Q_c for example (60) are those from section 4.2.2, including ‘Is Eve single?’, ‘Is Eve in a relation with Ann?’ and ‘Is Eve in a relation with Bea?’. Given what was said in section 4.2.2, (60b) will be defined according to (72) as the ordinary value of the embedded clause and its alternative based on **the person Eve danced with** resolve the first QuD in the same way and do not address the other ones differently. (60c) will be undefined: the ordinary value of the embedded clause and the alternative address the second and the third QUD differently.

The contrast between (73a) and (73b) in scenario 15 from (61), is derived in a similar way. We focus on the crucial case: **Eve is involved with a linguist** and **Eve is involved with a person she danced with**, the ordinary and the relevant replacement value for the embedded clause in (73a), are contextually equivalent, resolving ‘Is Eve in a relation with a linguist?’ in the same way in scenario 15. But in (73b), **Eve is involved with a physicist** and **Eve is involved with a person she danced with** are not contextually equivalent relative to ‘Is Eve in a relation with a physicist?’ and resolve it differently.

- | | | |
|------|--|-----------------|
| (73) | a. <i>Joe thinks Eve is involved with a linguist.</i> | ✓in scenario 15 |
| | b. <i>Joe thinks Eve is involved with a physicist.</i> | ✗in scenario 15 |

All of this hinges on the right questions being in the set of projected questions. For example, questions like ‘Is Eve dating Ann?’ and ‘Is Eve dating Bea?’ should be in that set together with ‘Is Eve single?’. But if ‘Is Eve single?’ – and this is crucial for our idea – is in this set, it is hard to see how the former questions can be excluded: they are intuitively natural follow-up questions if ‘Is Eve single?’ is resolved negatively. Existing theories of the structure of QuDs and context account for this by assuming that the intuitive follow-ups are sub-questions of the more general question (van Kuppevelt 1991, 1995, 1996, Roberts 1996, Büring 2003, Hénot-Mortier 2025).

However, we must add one more level of complexity to the QuD-constraint. The definedness condition in (72) requires there to be a projected question that is resolved by the content of the embedded clause (otherwise its first conjunct would not be met). It looks like this will have disastrous consequences. Take scenario 1a, a slight modification from scenario 1: all that has been changed is that the discourse is now about the belief state of the guests. (6b), repeated

in (74b) is still fine in scenario 1a. But don't we predict it to be out? After all, the embedded proposition, **Eve is involved with Ann** does not resolve a question about a belief state (for that, we would need something like **Bob thinks that Eve is involved with Ann**). We don't even need to appeal to replacement – (74) is a classical *de dicto* case – for this problem to occur.

- (74) a. SCENARIO 1A: Joe and Bob went to a party. Ann and Eve were among the guests. Bob knows both Ann and Eve well. He didn't see them together at the party. ... There is discussion about what the guests believe about Eve's relationship status. I say:
 b. *Bob thinks that Eve is involved with Ann.* ✓

We could of course assume that questions about belief states are usually tied to ones about the content of the belief (i.e. 'Does Bob think Eve is single? And is she?'). But examples like (75) in scenario 9 from (49), suggest this cannot be the general strategy – and that we must amend the QuD constraint so that it can also appeal to questions about the belief state *simpliciter*.

- (75) *Ha! Joe thinks that one of the winners lost.* ✓in scenario 9

The ordinary meaning of the embedded clause in (76) is the contradiction, which can never resolve a question in a context: the intuition behind (66biii) is that since the contradiction entails any proposition, it entails every possible answer to a question and thus cannot contribute to resolving it. Hence the QuD constraint as it is cannot be met and (75b) should be unacceptable.

This prediction would change if the whole sentence in (75b) were available for checking the QuD-constraint, i.e., if it included questions about attitude states. We thus make the QuD-condition contributed by *think* disjunctive as in (76) (with the new part underlined): for all alternatives p to the complement clause S of *think* there is a QuD Q such that either p and the normal value of S resolve the QuD in the same way (and do not address any of the projected questions differently) or the propositions saying that the subject s believes p and that s believes the normal value of S resolve the QuD in the same way (and do not address any of the projected questions differently).

- (76) $\mathcal{F}^{g,c,Q_c^S}(\text{think } S)(\langle a, \{a\} \rangle)$
 defined if $\forall p \in \llbracket S \rrbracket^t \rightarrow \exists Q \in Q_c^S [(p =_c^Q \llbracket S \rrbracket^o \wedge \forall Q' \in Q_c^S (p \sim_c^{Q'} \llbracket S \rrbracket^o)) \vee$
 $(\lambda w. \forall w' \in \text{DOX}_{a,w}(p(w')) =_c^Q \lambda w. \forall w' \in \text{DOX}_{a,w}(\llbracket S \rrbracket^o(w')) \wedge$
 $\forall Q' \in Q_c^S (\lambda w. \forall w' \in \text{DOX}_{a,w}(p(w')) \sim_c^{Q'} \lambda w. \forall w' \in \text{DOX}_{a,w}(\llbracket S \rrbracket^o(w')))]$
 if defined, $\mathcal{F}^{g,c,Q}(\text{think } S)(a) = \langle \lambda w. \exists p \in \llbracket S \rrbracket^t (\forall w' \in \text{DOX}_{a,w}(p(w')), \{ \lambda w. \exists p \in \llbracket S \rrbracket^t (\forall w' \in \text{DOX}_{a,w}(p(w')) \}$

Returning to (75b), the underlined part of (76) allows us to look at the propositions **Joe thinks one of the winners lost** – resulting from the ordinary value of the complement – and **Joe thinks that Ann, Bea, or Carl lost** – resulting from a replacement for **one of the winners**. Both negatively resolve the QuD 'Does Joe believe that Ann, Bea or Carl won?'.

4.3 Evidence for the QuD-constraint

The QuD-constraint makes predictions beyond the cases just discussed, as the context and the available QuDs should affect the accessibility of NDD-construals. We will look at these predictions for the phenomena discussed in sections 2 and 3,³⁴ but it will be helpful to first state the effects of (76) more schematically, using (77) as our abstract example.

- (77) $[\phi \text{ Joe thinks } [\psi \text{ that Eve is involved with Ann.}]]$

³⁴To the extent that QuDs different from those we will be assuming can be accommodated, judgments could improve.

The definedness condition in (76) can target both the embedded clause (ψ) and the matrix clause (ϕ), depending on the question involved. For questions Q in Q^c asking about ψ , we must check whether $\llbracket \psi \rrbracket^o$ and its alternatives are indistinguishable relative to Q , and for questions Q in Q^c asking about ϕ , we must check whether $\llbracket \phi \rrbracket^o$ and its alternatives are indistinguishable relative to Q . All propositions considered will be relativized to the utterance context c , i.e., conjoined with contextual information. When ψ is considered, we will enrich the propositions introduced by the embedded clause ‘directly’ with contextual information, which can have the effect of mimicking certain aspects of co-extensionality constraints for the purposes of definedness: if we know that Ann is the person Eve danced with, both the o -value of ψ , **Eve is involved with Ann**, and the alternatives, say, **Eve is involved with the person Eve danced with** can be conjoined with **Ann is the person Eve danced with**. So for the purposes of the QuD- constraint, the two resulting propositions would be ‘about’ the same individual. No such effect concerning the embedded content will emerge if ϕ is targeted: the contextual information **Ann is the person Eve danced with** will be conjoined with **Joe thinks Eve is involved with Ann** and alternatives like **Joe thinks Eve is involved with the person dancing with Eve**. As these involve an intensional component, adding contextual information will not let us infer for the purposes of the QuD-constraint that **Ann** and **the person Eve danced with** pick out the same individual in the context worlds.

Since the QuD-constraint appeals to a set of projected questions (rather than a single question) it stands to reason that we will usually consider both questions related to ϕ and questions related to ψ . This will guarantee that contextual information can ‘access’ the embedded content for comparison purposes (and thus, by connecting the ordinary and the replacement meaning, potentially rein in replacement in terms of reference in the context worlds).³⁵ But if $\llbracket \psi \rrbracket^o$ is a contradiction (as in scenario 9 in (49)) or entailed by the context, $\llbracket \psi \rrbracket$ will not be able to resolve a question.

4.3.1 The QuD-constraint in *de re*

We saw that (9) is fine in scenario 1, both repeated as (78). The explanation for why the QUD constraint is satisfied for (78b) in scenario 1 is parallel to the one for (60b) in scenario 14.

- (78) a. SCENARIO 1: Joe and Bob went to a party. Ann and Eve were among the guests. ... Joe recognizes Eve, but not Ann and does not think Eve knows Ann. He saw them dancing with each other and thinks Eve and the person she was dancing with are dating. There is discussion at the party whether Eve is single.
 b. *Joe thinks that Eve is involved with Ann.* ✓

Consider now the minimal modification of the scenario in (79a). (79b) is degraded in scenario 1b. This is surprising on standard accounts of *de re*. After all (79d) holds in scenario 1b.³⁶

- (79) a. SCENARIO 1B: [...] We are discussing whether Joe thinks that Ann was at the party.
 b. [ϕ *Joe knows/thinks* [ψ that *Ann was at the party*]] ✗
 c. **Joe thinks/knows Ann was at the party** ordinary value of ψ
 d. **Joe think/knows the person Eve was dancing with was at the party** replacement of ψ

The QuD condition helps us make sense of this. The question to be resolved must concern Joe’s belief state (ϕ): if we tried to ask about the embedded clause ψ (‘Was Ann at the party?’), neither the ordinary meaning **Ann was at the party** nor the replacement proposition **the person dancing with Eve was at the party** together with the context resolve this question according to (66a) as

³⁵ Aonuki and Davidson 2025 show that both questions targeting the belief content and those targeting the attitude license NDD-construals, however, they do not consider the option of projected questions.

³⁶ The factive presupposition of *know* regarding **the person dancing with Eve was at the party** is satisfied here.

the context already entails the answer (asking such a question in this context would be odd). So the first conjunct of the QUD-constraint would be violated. As only a QUD about Joe's belief state ('Does Joe think Ann was at the party?') can be resolved in scenario 1b, the underlined part of (76) is relevant: we will consider the propositions (79c), and (79d) – so those propositions, rather than the embedded propositions on their own will be relativized to the context. Since (79c) and (79d) involve an intensional component, the contextual information that Ann is the person dancing with Eve will not help in establishing contextual equivalence relative to the QUD. While the ordinary value of ϕ , (79c), resolves the question positively, (79d) does not address it, so the two address the question differently and the QUD-constraint blocks replacement.³⁷

Matters change if it is immaterial for the question whether **Ann** and **the person Eve was dancing with** pick out the same individual in the context worlds: in the minimal modification of scenario 1 in (80a) the salient QUD to be resolved is again about the belief-state as a whole (ϕ), but now it is asking if Joe believes that Eve is single. The sentence in (80b) is acceptable.

- (80) a. SCENARIO 1c: Joe and Bob went to a party. Ann and Eve were among the guests. ...Joe recognizes Eve, but not Ann and does not think Eve knows Ann. He saw them dancing with each other and thinks Eve and the person she was dancing with are dating. There is discussion at the party whether Joe thinks Eve is single.
- b. [ϕ Joe thinks [ψ that Eve is involved with Ann]] ✓
- c. **Joe thinks Eve is involved with Ann** ordinary value of ψ
- d. **Joe thinks Eve is involved with the person dancing with Eve** replacement of ψ

The QUD-condition in (76) predicts this. The proposition (80c) resolves the QUD positively, and so does the proposition (80d) resulting from replacement.

Contrast this with the further modification in (81a). Again, the QUD to be resolved is about the belief state (ϕ). The minimally different sentence in (81b) is unacceptable. Again, this is predicted: the proposition (81c), based on the ordinary value resolves the QUD ('Does Joe think Ann is single?') positively, but the replacement based (81d) does not address this QUD.

- (81) a. SCENARIO 1d: [...] We are discussing whether Joe thinks that Ann is single
- b. [ϕ Joe thinks [ψ that Ann is involved with Eve]] ✗
- c. **Joe thinks Ann is involved with Ann** ordinary value of ψ
- d. **Joe thinks the person dancing with Eve is involved with Eve** replacement of ψ

But this reasoning raises an issue. As outlined above, if none of the questions in the set of projected questions targets the embedded clause (ψ), contextual information cannot 'enrich' the embedded content directly. We appealed to this property to explain the unacceptability of (79b)

³⁷In this light, consider again (i) in scenario 8 from (48). Here the QUD is 'Has Ann has arrived?'. The ordinary value of ψ'' resolves this question positively. **flight AF62 has landed**, based on replacement of **Ann** with **AF62**, as suggested above, also does so together with the contextual information that Ann is a passenger on flight AF62.

- (i) [ϕ'' Joe thinks [ψ'' that Ann has arrived]] ✓in scenario 8

If the question were about Joe's belief state, which we could force if Ann's having arrived is entailed by the context, we would not enrich the ordinary meaning and the replacement of ψ'' with contextual information, but the propositions including the attitude (**Joe thinks that Ann has arrived** and **Joe thinks that AF 62 has landed**). Thus the latter would include no information about Ann having landed. This correctly predicts that (iib) is bad in scenario (iia).

- (ii) a. SCENARIO: Ann was on flight AF62. She texted Bea and Cate that she arrived. Bea talks to their friend Joe, who works at the airport and knows the flight schedule, not that Ann is traveling. Bea doesn't mention Ann. Joe says AF 62 has landed. Then Bea and Cate discuss who they should alert to Ann being in town. Bea, to Cate: *Well,...*
- b. [ϕ'' Joe thinks/knows [ψ'' that Ann has arrived]] ✗

in scenario 1b. But we also used contextual enrichment of the embedded clause to ensure that individual concepts yield the same individual in the context worlds in (60). We should thus lose this restriction in those contexts where the projected questions are solely about the belief state (ϕ).

In light of this problem, consider first scenario 1e, which is like scenario 1c, except that Bea is contextually salient. While (80b) is still acceptable, (82b) is bad. At first, this contrast is unexpected: the QuD ‘Does Joe think Eve is single?’ is resolved positively by both the ordinary and the replacement value of (80b) ((79c) and (79d)), but is also resolved positively by both values of (82b), (82c) and (82d)) – so no difference here. However, the QuD-constraint does not only force the ordinary and the replacement proposition to resolve a question in Q in the same way, it also imposes a constraint regarding all other projected questions – that they cannot be addressed differently by the ordinary and the replacement proposition. Arguably, the questions ‘Is Eve involved with Ann?’ and ‘Is Eve involved with Bea?’, which relate to ψ , are in this set for (80b) and (82b), respectively, and – in analogy to the discussion in section 4.2.2 – are addressed in the same way by (79c) and (79d), but not addressed in the same way by (82c) and (82d). In other words, as projected questions concerning the embedded content, while not being the main point of interest, are still possible in scenarios 1b and 1d, ‘random’ replacement is blocked.

- (82) a. SCENARIO 1D ... discussion that Bea is at the party and if Joe thinks Eve is single.
b. *Joe thinks that Eve is involved with Bea.* ✗
c. **Joe thinks Eve is involved with Bea** ordinary value of ψ
d. **Joe thinks Eve is involved with the person dancing with Eve** replacement of ψ

But what if the context prevents any such questions being at stake, by entailing all of their answers, as in the slight modification of scenario 1b in 1e?

- (83) a. SCENARIO 1E [...] We all know that Eve isn’t dating Ann, and that she isn’t dating at all, but we are discussing whether Joe thinks she is. We are also discussing that Bea was at the party.
b. [ϕ *Joe thinks [ψ that Ann is involved with Eve]*] ?
c. [ϕ *Joe thinks [ψ that Bea is involved with Eve]*] ✗

(83c) is still out in scenario 1e, and although (83b) is slightly worse in scenario 1e than in scenario 1b, the contrast between (83c) and (83b) seems unexpected: questions about the embedded content (ψ) cannot be raised, hence our constraints on ‘random’ replacement should not be observable.

However, there is a difference between the question that must be *resolved* in the same way by the ordinary value and the replacement proposition, and those that simply must not be addressed differently. In a context like 1e, the former can only be a question about the belief state (ϕ). But as long as there is another projected question that doesn’t involve an intensional component, and which for which **Bea** and **the person dancing with Eve** will yield propositions that don’t address this question in the same way, we will constrain replacement sufficiently, resulting in (83b) being bad. As there are a large number of these questions (targeting properties not shared by **Bea** and the value of **the person dancing with Eve** in the context), this situation cannot be excluded.

4.3.2 QUD-constraint on NT and transparent construals of verbal predicates

The pattern for NT-construals and our explanations for it are fully parallel. Recall that (84b) is acceptable in scenario 4 (= (39)). The reasoning here is analogous to that for (78): the QuD to be resolved concerns the embedded clause (‘Is Eve in a relationship?’) so the ordinary value of the embedded clause, (84c), and the replacement, (84d) can be conjoined with the contextual information that Ann, Bea and Cate are linguists. Both propositions resolve the QuD positively.

- (84) a. SCENARIO 4: Joe and Bob went to a party. ... Joe thought all guests were biologists. Except for three linguists called Ann, Bea, and Cate, this was the case. Joe doesn't know them or their names. He saw each of them dancing with Eve, who he knows. So he thinks Eve and one of them are lovers but is not sure which of them. We are discussing whether Eve is in a relationship.
- b. [_{ϕ} Joe thinks [_{ψ} Eve is involved with a linguist]]
- c. **Eve is involved with a linguist** ordinary
- d. **Eve is involved with a person dancing with Eve** replacement

In the slightly modified scenario (85a), sentence (85b) is unacceptable. This case is analogous to scenario (80) and again, the judgment is unexpected on existing approaches as predicted meaning in (85c) holds in scenario 4a. On our view, we expect the observed degradedness. The reasoning is parallel to that for (80): the QuD to be resolved must be about Joe's belief state ('Does Joe believe a linguist was at the party?'), the ordinary meaning, (85c), resolves this question positively, but the replacement proposition, (85d), does not address it, thus violating the QuD-constraint.

- (85) a. SCENARIO 4A: [...] We are discussing whether Joe thinks a linguist was at the party.
- b. [_{ϕ} Joe knows/thinks [_{ψ} a linguist was at the party]] ✗
- c. **Joe thinks/knows a linguist was a the party** ordinary
- d. **Joe thinks/knows a person dancing with Eve was a the party** replacement

And as predicted, once the QuD to be resolved can address ψ as in scenario 4b, (85b) becomes acceptable as in (86): both the ordinary value, (86c) and the replacement value, (86d) can be directly supplemented with contextual information (all pink-haired women are linguists) and the resulting propositions resolve the QuD ('Was there a linguist at the party?') in the same way.

- (86) a. SCENARIO 4B: Joe went to a party. Eve was there, and he recognized her. He saw Eve dancing with three pink-haired women, who he didn't know. He tells me that he thinks that Eve and one of the pink-haired women are lovers but is not sure which of them. I know that there are only three pink-haired women, and they are all linguists. We are discussing whether there were any linguists at the party. I say:
- b. *Joe thinks a linguist was at the party.* ✓
- c. **a linguist was at the party** ordinary
- d. **a pink-haired woman was at the party** replacement

Turning to transparent construals verbal predicates, it has been observed in the literature that they are tied to contextual factors (Fodor (1970), Schwager (2011), Sudo (2014)): Schwager suggests that a verbal predicate P can be replaced by a predicate Q if P and Q are contextually equivalent.³⁸ Our QuD constraint can be seen as a generalization of this idea, comparing, however, propositions instead of predicates. We have already seen the effect of the QuD: In (87) (=53), the scenario makes 'Are any of Eve's partners Canadian?' salient. The ordinary value of ψ resolves this question positively as does the replacement proposition **the person Eve danced with is Canadian** combined with the contextual information that the dancer is Ann, a Canadian.³⁹

³⁸She includes a prerequisite so as to deal with predicates with empty extensions in the utterance world. See fn ??.

³⁹(i) is another example. 'Does the boss spend his work time on hobbies?' is salient in the scenario. Both the ordinary value of the embedded clause and the alternative with **went skiing** replaced by **went hiking** resolve it positively if assumptions from the context are included. A transparent construal of the VP is thus correctly predicted to be licensed.

- (i) a. SCENARIO: The staff discusses how some managers of the company they are working for might be spending all their time on hobbies instead of work. Ann thinks Joe is a manager. He might be but we are not entirely sure. Joe went skiing during work hours taking off from the company's premisses and Ann saw him leave. She falsely concluded he went hiking given that he was wearing a thick jacket but did not

- (87) a. SCENARIO 11: Joe went to a party. He thought all guests were American. Except for one Canadian called Ann this is true. He saw Eve dancing with exactly one person and thinks Eve and the person he saw her dancing with are dating. The person he saw Eve dancing with happens to be Ann. Joe neither knows Ann nor her name. He also was not able to see Ann's face. We know that Eve changes partners constantly and has multiple ones at any point. Not being sure who her current partners are, we are wondering where they might come from. I say:
- b. [ϕ Joe thinks [ψ that one of Eve's partners is Canadian]] ✓

The scenario in Percus's 2000 original example in (88) (= (52)), however, where a transparent construal did not seem accessible, is so impoverished that no QuD can easily be accommodated, so no non-trivial replacements are not (easily) available.

- (88) a. SCENARIO 10: Ann mistakenly believes that Bert is Joe's brother. She moreover mistakenly believes that Bert is American, while in fact he is Canadian.
- b. *Ann thinks that Joe's brother is Canadian.* ✗

4.3.3 QUD-constraint with WO

Consider the scenario in (89a) (from (55)) and the minimally different one in (89b) (a continuation of scenario 12). (89c) is acceptable in scenario 12 (as discussed above), but not in scenario 12a.

- (89) a. SCENARIO 12: Abe falsely thinks his neighborhood is full of terrorists. He goes to the police. Officer Bea shows him photos of people. For each photo, she asks if the person on it is a terrorist and if they live on Abe's street. When done, Abe is asked if there are terrorists in the neighborhood who are not on any of the photos, if he knows how many people he identified as terrorists and how many of those as living on his street. Abe says no and claims he only knows he identified several as terrorists living on his street. Bea knows that Abe has identified 11 as terrorists in total and 7 of these as living on his street. Bea tells her colleague, who, like Bea, is worried about Abe going crazy. Bea:
- b. SCENARIO 12A: [...] Bea and her colleagues know that Abe does not mind terrorists living in his neighborhood as long as the majority of them does not live on his street, which would make him move. A colleague asks Bea: Will Abe have to move? Bea:
- c. [ϕ Abe believes [ψ that most terrorists live on his street]] scen. 13: ✓ scen 13a: ✗

This contrast follows from the QUD constraint. The QuD in (89a) is about Abe's belief state (ϕ), something like 'Does Abe think that people he identified as terrorists live on his street?'. The ordinary value in (90a) as well as the replacement proposition in (90b) answer this positively. In contrast, the two propositions do not resolve the QuD in (89b) in the same way ('Are Abe's beliefs such that he will move?'): (90a) would resolve it positively, (90b) would not.

- (90) a. **Abe believes that most terrorists live on his street.** ordinary
- b. **Abe believes that several terrorists live on his street.** replacement
-
- b. have any skiing gear with him. I know all this and say: *Trust in the management has eroded...*
- b. *Ann even believes that a manager went skiing during work hours.* ✓

4.4 Summary: constraints on replacement

We suggested that replacement (or underspecification) is generally constrained by ‘discourse stability’, a pragmatic mechanism ensuring that the ordinary meaning and its replacements ‘behave the same’ for the purposes of the projected discourse. In our more modest implementation, we treated the ‘projected discourse’ as a set of projected questions, i.e., questions addressed by the sentence uttered or its parts and likely follow-up questions. We encoded the constraint itself as a presupposition tied to intensional operators; it compares the ordinary meanings of parts of the sentence with their replacements, checking (i) if there is a projected question that they resolve in the same way (the question the utterance can be thought to answer) and (ii) whether they ‘behave the same’ relative to all other projected questions (those that might be or might have been entertained; the ordinary and the replacement meaning cannot answer them differently). For the purpose of this comparison (which essentially checks whether a given replacement can be used without misleading the hearer), the meanings are relativized to the context. Thus, if the set contains questions that concern the embedded clause, contextual information can enrich the ordinary and replacement meanings of the latter, adding information on e.g., contextual equivalence that is not part of the information state of the attitude holder.⁴⁰

⁴⁰In light of all this, consider the ‘double vision problem’ (Quine 1956). The literature reports that (ib) and (ic) can both be true in scenario (ia) (all adapted from Quine 1956). The intuition followed by Quine (a.o.) is that (ib) is true in virtue of the fact that Ralph believes of the man in the brown hat that he is a spy, with Orcutt as the res, while (ic) is true in virtue of Ralph not believing the man on the beach is a spy, again with Orcutt as the res. Our approach derives this case follows: the *t*-values for the embedded clauses differ as in (ii). It is consistent for Ralph to both believe one of the propositions in (ia) and one of those in (ib) (as long as they are not **Orcutt is a spy** and **Orcutt is not a spy**).

- (i) a. SCENARIO: There is a man in a brown hat whom Ralph has glimpsed several times under questionable circumstances; Ralph suspects he is a spy. Also there is a grey-haired man, vaguely known to Ralph as a pillar of the community, whom Ralph is not aware of having seen except once at the beach. Ralph does not know that the men are one and the same Bernard Orcutt.
- b. *Ralph believes that Orcutt is a spy.* ✓
- c. *Ralph believes that Orcutt is not a spy.* ✓
- (ii) a. $\llbracket \text{that Orcutt is a spy} \rrbracket' = \{ \text{Orcutt is a spy, the man in the brown hat is a spy, the man on the beach is a spy, ...} \}$
- b. $\llbracket \text{that Orcutt is not a spy} \rrbracket' = \{ \text{Orcutt is not a spy, the man in the brown hat is not a spy, the man on the beach is not a spy, ...} \}$

(iiia) is expected to be fine in scenario (ia) for the reason just outlined. In (iiib), negation takes scope over *believe* in the second conjunct. As *believe* contributes existential quantification over alternatives, the second conjunct will require that Ralph believes no proposition in (iia). This contradicts the first conjunct, so we explain the contrast in (iii).

- (iii) a. *Ralph believes that Orcutt is a spy, and he believes that Orcutt is not a spy.* ✓ in scenario (ia)
- b. *Ralph believes that Orcutt is a spy, and he doesn't believe that Orcutt is a spy.* ?? in scenario (ia)

To the extent that (iiib) is acceptable at all in scenario (ia), our account requires the two conjuncts to address different QUDs, i.e., a context change between the conjuncts (see e.g. von Stechow 2001 for related discussion). (Another option which we don't go into here would be to appeal to neg-raising.) Such a context change might also underlie an observation connected to the parallelism data from section 2.2: Our examples with quantifiers lead us to expect that parallelism is also observable with ellipsis, namely, that (ivb) should be fully acceptable in our original scenario 1. Yet the judgments are more varied than in the quantifier cases in section 2.2. Crucially, however, there is a contrast in acceptability between (iva) and (ivb). Our account helps us get a grasp on this: (ivb) is predicted to be false in scenario 1, for the same reason that the sentences with *only* are. (iva) is predicted to be true, i.e., we correctly expect it to be more acceptable than (ivb). The relative degradedness of (iva) intuited by some speakers could thus be due to independent factors: As opposed to the examples with quantifiers, (iva) consists of two syntactically expressed conjuncts. Each of these could resolve a different immediate QUD, which could negatively affect parallelism.

- (iv) a. *Joe thinks that Eve is involved with Ann, and Bob does too.* ✓ in scenario 1
- b. *Joe thinks that Eve is involved with Ann, but Bob doesn't.* ✗ in scenario 1

Our way of restricting replacement departs significantly from the literature: As made explicit above we do not appeal to relations between extensions in the evaluation world. We also do not appeal to acquaintance, which, starting with Kaplan (1968), has often been taken to constrain replacement in *de re* (Sharvit 1998, Percus and Sauerland 2003, Ninan 2008, 2012, Maier 2009 a.m.o.): The idea is that for replacement to be possible, the attitude subject *a* must be acquainted with the value of the individual concept **f** (the res) in the actual world and the individual *a* take themselves to be in their attitude alternatives *w* must be acquainted with the value of **f** in *w*.

Acquaintance was originally motivated by the shortest-spy problem. (91b) is unacceptable in scenario 17. But if we were allowed to replace **Mia** with **the shortest spy**, (91b) should be true since Ann believes that the shortest spy is a spy. The standard reaction is to say that what goes wrong is that Ann is not acquainted with Mia and consequently replacement is blocked.

- (91) a. SCENARIO 1: Ann believes there is a shortest spy. Mia, who Ann is not acquainted with, is in fact the shortest spy.
- b. [ϕ Ann knows/believes [ψ that Mia is a spy]] **X**
- (i) **Mia is a spy** ordinary ψ
- (ii) **the shortest spy is a spy** replacement ψ
- c. (i) **Ann thinks Mia is a spy** ordinary ϕ
- (ii) **Ann thinks the shortest spy is a spy** replacement ϕ

However, our account derives this judgment without appealing to acquaintance.⁴¹ In order for the sentence to be true, we need to appeal to replacement, but the question to be resolved cannot be tied to the embedded clause ϕ (e.g., ‘Is Mia a spy?’): If the ordinary value is entailed by the context, the question cannot be raised. If it is not entailed by the context, then the ordinary value of ϕ , (91bi), and the replacement value, (91bii), will not resolve the question the same way according to (66) because (91bii) is the tautology. Potential questions concerning the entire sentence (ϕ), e.g., ‘Does Ann think Mia is a spy?’, won’t be resolved in the same way by the ordinary value (91ci) and the replacement value (91cii) either, because again the latter is the tautology. In other words, a replacement value that would yield the tautology can never be used.⁴²

As a final point, let us consider whether our account fares concerning Schwager’s 2011 problem for theories of NDD-construals appealing to relations between extensions in the actual world. Schwager notes that (92b) seems true in scenario (92a) in virtue of a transparent construal of *building with 192 floors in NY*. Thus standard accounts of NT predict that the meaning for (92b) should say that in all of Abe’s belief worlds the proposition [λw .there is an *x* s. th. *x* is a building with 192 floors in NY in w^* & Abe visited *x* in *w*] holds. But **building with 192 floors in NY** has an empty extension in the actual world w^* , so this is the contradiction.⁴³

⁴¹The acquaintance condition has been challenged independently (Bonomi 1995, Ninan 2008, 2012, Hawthorne and Manley 2012 a.o., but cf. Sharvit 1998): In scenario (ia) Joe is not acquainted with Roy, but replacing **Roy** with **Ann’s lover** is possible. Under our account, a QuD like ‘Is Ann’s lover a fool?’ is resolved in the same way by the ordinary and the replacement meanings of the embedded clause (**Roy is a fool** and **Ann’s lover is a fool** and) together with contextual information (**Roy is Ann’s lover**), so we correctly predict (91b) to be fine in scenario (91a).

- (i) a. SCENARIO: Joe thinks Ann has a lover. He says to Bob: “Ann looks happy. She must have a lover. I don’t know who he is, I have never seen him, but I think he is a fool.” Bob knows Roy is Ann’s lover. Joe does not know and has never heard of Roy. Bob:
- b. *Joe believes that Roy is a fool.* ✓ (adapted from Sharvit 1998)

⁴²The same holds for a replacement that would yield contradiction. This case is distinct from the one where the ordinary value yields the contradiction, as in scenario 9 in (49), and also observable in *Ann doesn’t think the shortest spy is a spy* which is fine in contexts where we believe Mia is the shortest spy and Ann doesn’t believe Mia is a spy.

⁴³(i) raises a similar issue for *de re*: As there are no superheroes in the actual world, **Superman** and **superhero called Clark Kent...** are undefined, so the co-extensionality condition of standard *de re* theories cannot be met.

- (92) a. SCENARIO 18: Abe is looking at the Burj Khalifa. Not knowing that it has 191 floors and is the world's tallest building, he thinks: 'The Empire State building I visited in New York is one floor taller'.
 b. *Abe thinks he visited a building with 192 floors in New York.* ✓
 (adapted from Schwager 2011)

Our account does not seem to run into this issue as replacement does not require co-extensionality or overlap of properties in the actual world. Yet this only holds if the QuD to be resolved targets the entire belief state. If it targeted the embedded clause, as in (93), we would face a problem: as there is no building with 192 floors in the utterance context, the intersections in (93b) and (93c) amount to contradictions and thus do not resolve the question. (91b) should be degraded.

- (93) a. Salient QUD: *Did Abe visit a building with 192 floors in New York?*
 b. $c \cap \lambda w. \text{Abe visited a building with 192 floors in New York in } w = \emptyset$
 c. $c \cap \lambda w. \text{Abe visited a building in NY that is one floor taller than the BK in } w = \emptyset$

It is unclear to us whether a QuD like (93) is ever compatible with the sentence being judged true. But if so, we would have to state the QuD-constraint with the option of revised contexts, an variant we only allude to here: For the QuD in (93) to be resolved in the same way by the ordinary value of the embedded clause and its alternative based on the property **building in NY that is one floor taller than the BK** in a context c , there must be a world in c where Abe visited such a building. Adapting suggestions by Heim 1992 a revision of a context c for a proposition p could then be thought of as follows: If c contains p -worlds, its revision is identical to c (counterfactual worlds will only be considered if they have to). If c does not contain p -worlds, its revision amounts to a superset X of c containing a p -world, where X must satisfy all the presuppositions satisfied in c . The QUD-constraint could then makes reference to revisions of c for the ordinary value of the embedded clause S . As the revision of c would be vacuous as long as c contains worlds where that value is true, this would only have an effect for (92) and similar cases.

5 Conclusion and outlook

We proposed a novel perspective on NDD-construals which departs substantially from existing proposals in a number of respects. Partially building on Mayr and Schmitt 2024, we argued (i) that all NDD-construals are a uniform phenomenon and should have the same explanation (rather than being different phenomena resulting from different mechanisms), (ii) that this explanation should not appeal to specialized grammatical devices, but that NDD-phenomena reveal a property of meaning assignment *per se*, namely, underspecification, and (iii) that this underspecification is constrained by a notion of discourse stability, which we implemented as a condition that underspecification is only possible if all the meaning candidates 'behaving the same' in terms of the questions projected to come up in the discourse.

Our proposal thus clusters NDD-construals with phenomena that have previously been argued to involve some form of underspecification, such as non-maximality or imprecise meaning of round numerals. It should be clear, however, that our account does not yet provide a unification. As we laid out above, it seems plausible that underspecification is reined in by some version of Gricean reasoning appealing to principles like quantity and quality, but it is not straightforward

- (i) a. SCENARIO: Ann is confused. She saw a Superman movie. Now she cannot tell fact from fiction and is convinced she is married to the superhero who is Clark Kent in everyday life. Moreover she is not sure if Clark Kent is Superman, Batman, or Spiderman:
 b. *Ann thinks that she is married to Superman.* ✓

how make this intuition more concrete. The notion of ‘discourse stability’ could in the right direction – speakers try to not lead hearers too far astray concerning the projected discourse and hearers count on this – but we have not given a general implementation of this idea. Our rendering cannot be the full picture: it is unclear whether the projected discourse should indeed be captured via a question-based parameter; we arguably need a more localized application of the constraint (potentially targeting sub-clausal constituents) and, crucially, the mechanism should expand to underspecification in unembedded contexts (without intensional operators). The latter step will be essential for a general theory of underspecification that includes non-maximality, imprecision etc. Apart from these more conceptual issues there is also a question concerning the implementation: our use of alternatives is nothing but a technical remedy to make properties of the system visible.

A number of open questions concern NDD-construals more narrowly. For example, since our account assumes that NDD-construals result from properties of meaning *per se*, rather than dedicated grammatical devices, we will have to attribute all asymmetries concerning the availability of NDD-construals to the QuD-constraint or to operations targeting transparent values directly. This means, for instance, that the observation that transparent construals in the restrictor of DPs are more easily available than those for VPs should be derived in terms of asymmetries of how these different elements interact with the discourse and questions addressed in it. It also means that we would have to reassess those data points where a certain interaction between morphological marking and the availability of NDD-construals has been argued to be observable. While to our knowledge there don’t seem to be dedicated formal strategies that consistently correlate with the presence or absence of NDD-construals (an observation which fits our claim that no dedicated grammatical mechanisms are responsible for NDD-construals), certain morphological markers, such as subjunctive morphology in Romance, have been argued to suppress NDD-construals (e.g., Romero 1999 for Spanish, see also Quine 1956, Partee 2006, Recanati 2019 for related data). From our perspective, this could either mean that these configurations do not easily permit the types of questions needed to license the desired replacement or else that the morphemes that exhibit this type of behavior themselves target transparency value, i.e., block underspecification.

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